

ECSE 4560/ 6560

Modern Communication Systems

Lecture ~~5~~ 6: Wireless Systems Overview

Part 5 - Channel

Ish Jain, Fall 2025



Announcements

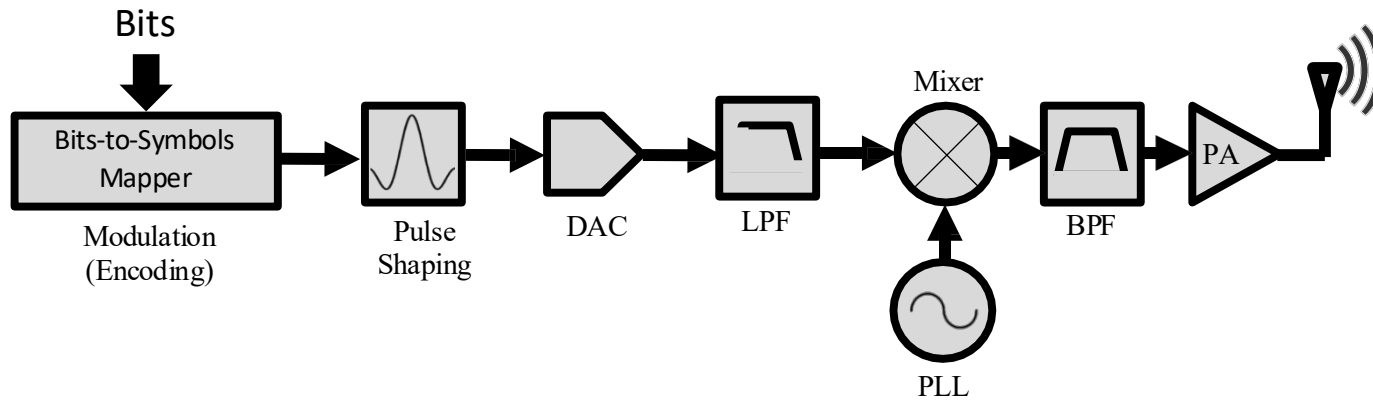
- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **Channel**
- Projects
 - A list of papers will be posted
 - Group formation [MCS Fall25 project.xlsx](#)
 - Proposal due 9/26
- HW1/Lab1 due Monday 9/22
 - Submit on Gradescope – Mark pages for each question parts
 - Matlab coding exercises – “Use Publish option to submit your code”
- HW2/Lab2 to be posted this weekend, due Mon 9/29

Today's Class: Wireless Overview 5

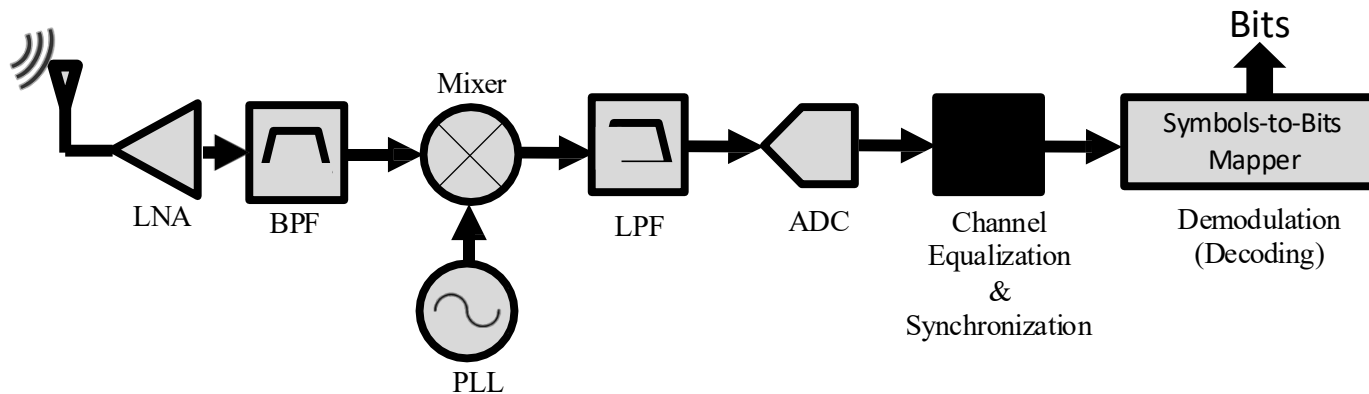
- QPSK BER (Review)
- Ideal Sampling and Reconstruction (Review)
- Practical ADC/DAC Sampling and reconstruction
- Wireless Channel

Wireless Communication System

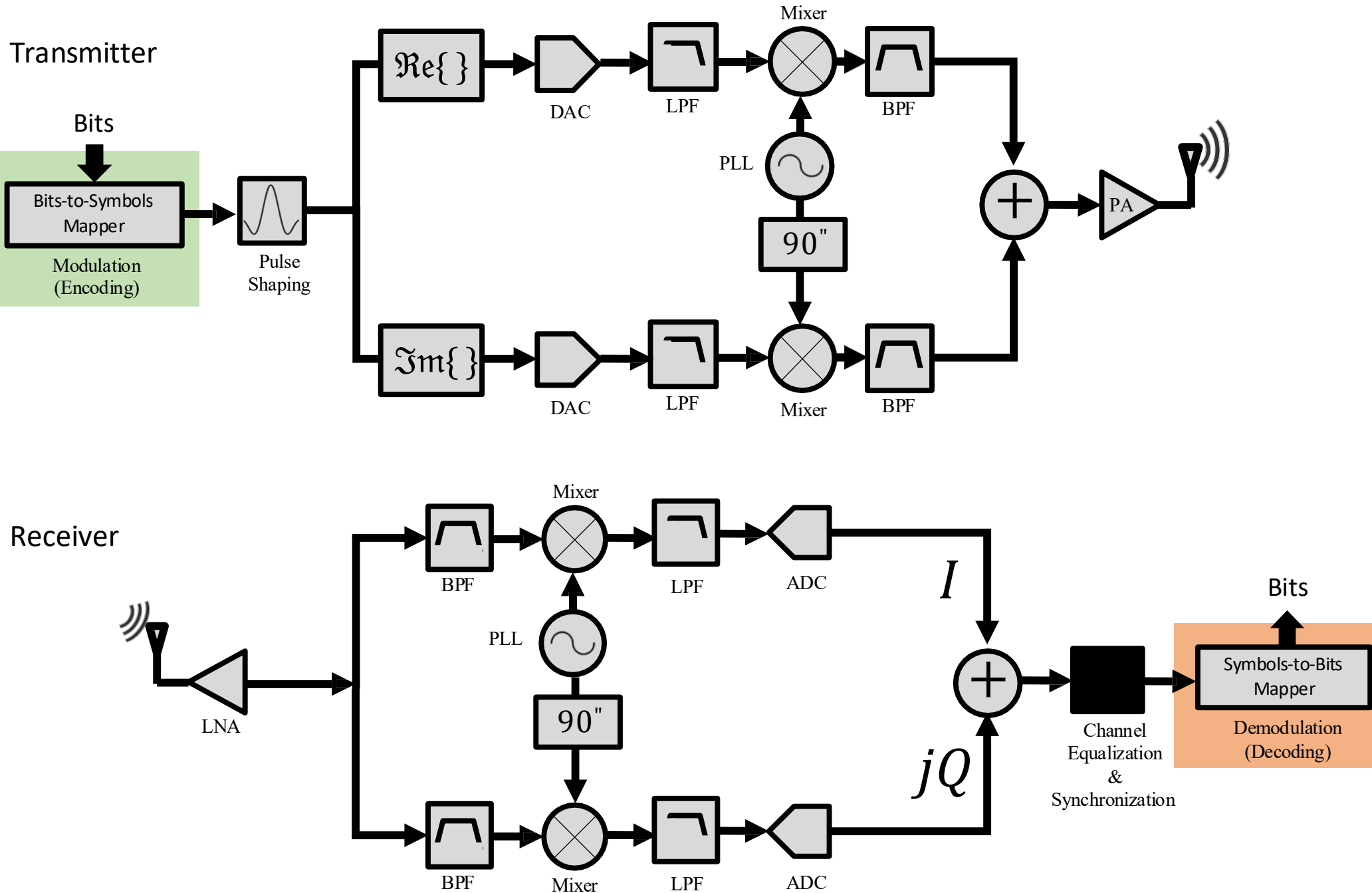
Transmitter



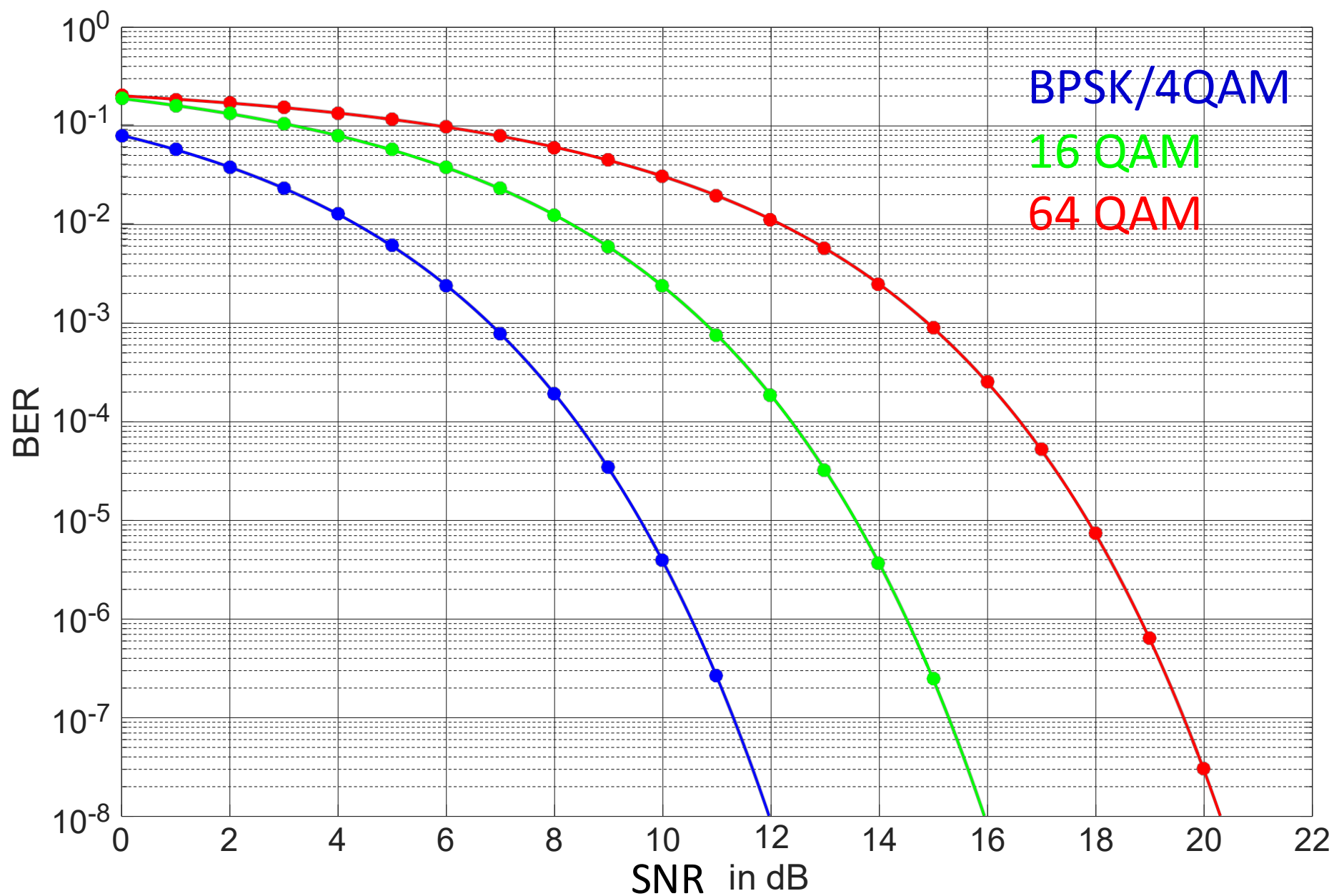
Receiver



Wireless Communication System



Bit Error Rate (BER)



Bit Error Rate (BER)

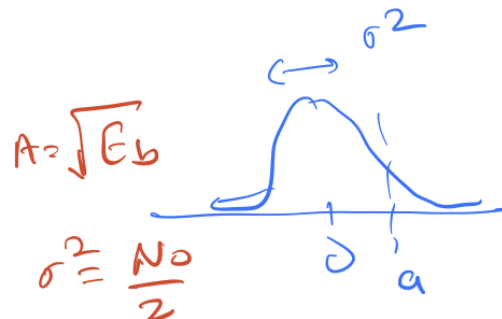
$$\begin{aligned}
 & P(\text{bit error}) \quad (y_i = -A + v) \\
 & = P(y_i > 0 \mid H_0) \text{ or } P(y_i < 0 \mid H_1) \\
 & = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2} \frac{(y_i - -A)^2}{\sigma^2}} = P(y_i \mid -A)
 \end{aligned}$$

$$\begin{aligned}
 & P(-A + v > 0) \\
 & : P(v > A) = P\left(\frac{v}{\sqrt{\sigma^2}} > \frac{A}{\sqrt{\sigma^2}}\right)
 \end{aligned}$$

$$\frac{v}{\sqrt{\sigma^2}} \sim N(0, 1)$$

$$\begin{aligned}
 & Q \text{ function} = P(\text{st-normal dist} > a) \\
 & \quad \underline{Q(a)}
 \end{aligned}$$

$$\begin{aligned}
 & P(v > A) \\
 & = Q\left(\frac{A}{\sqrt{\sigma^2}}\right) \\
 & = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) \quad \text{where } \frac{E_b}{N_0} = \text{SNR}
 \end{aligned}$$



Symbol Error Rate (SER)

16 QAM = $2^4 = 4$ ~~8~~ bits/symbol

bit error = P_b for each bit

what is symbol error rate?

SER

P_s = Prob that ~~any~~ one bit in error
at least

1 - no bits in error

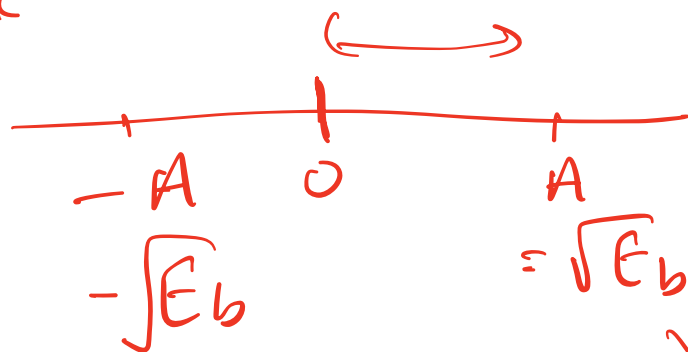
= 1 - bit 1 no error $(1 - P_b)$
x bit 2 no error $(1 - P_b)$

= $1 - (1 - P_b)^k$ k^{th} order mod.

$P_s \approx 2 P_b$ for $k=2$

BER for BPSK & QPSK

BPSK



$$E_b = A^2$$

$$A = \sqrt{E_b}$$

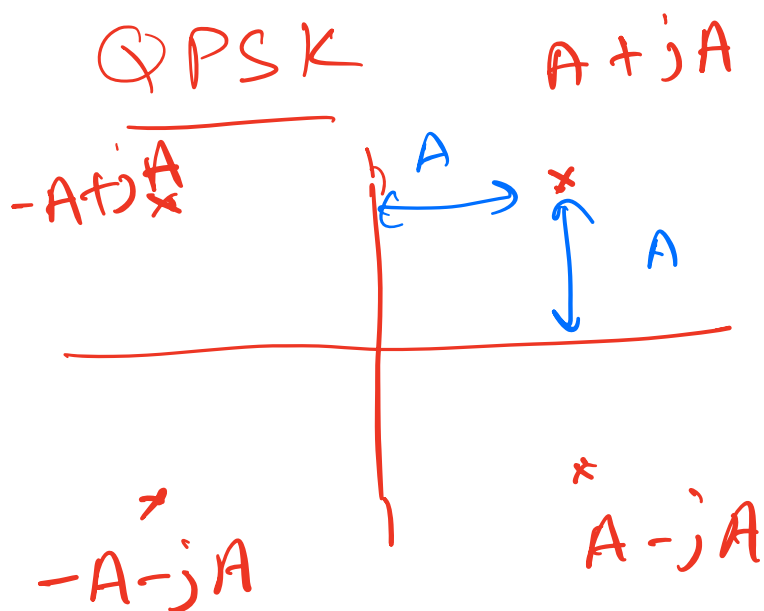
$$P_e = P(\gamma > \sqrt{E_b})$$

$$= P\left(\frac{\gamma}{\sqrt{\sigma^2}} > \sqrt{\frac{E_b}{\sigma^2}}\right)$$

$$\sigma^2 = \frac{N_0}{2}$$

$$= Q\left(\sqrt{\frac{E_b}{\sigma^2}}\right) = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

QPSK



$$E_s = [A]^2 + [jA]^2$$

$$= 2A^2$$

$$E_b = \frac{E_s}{2}$$

$$= A^2$$

$$P_e = P(\gamma > A) = P(\gamma > \sqrt{E_b})$$

$$= Q\sqrt{\frac{2E_b}{N_0}}$$

QPSK

$$P_e = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

bit
error
rate

$$P_s \approx 2 Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

symbol
error
rate

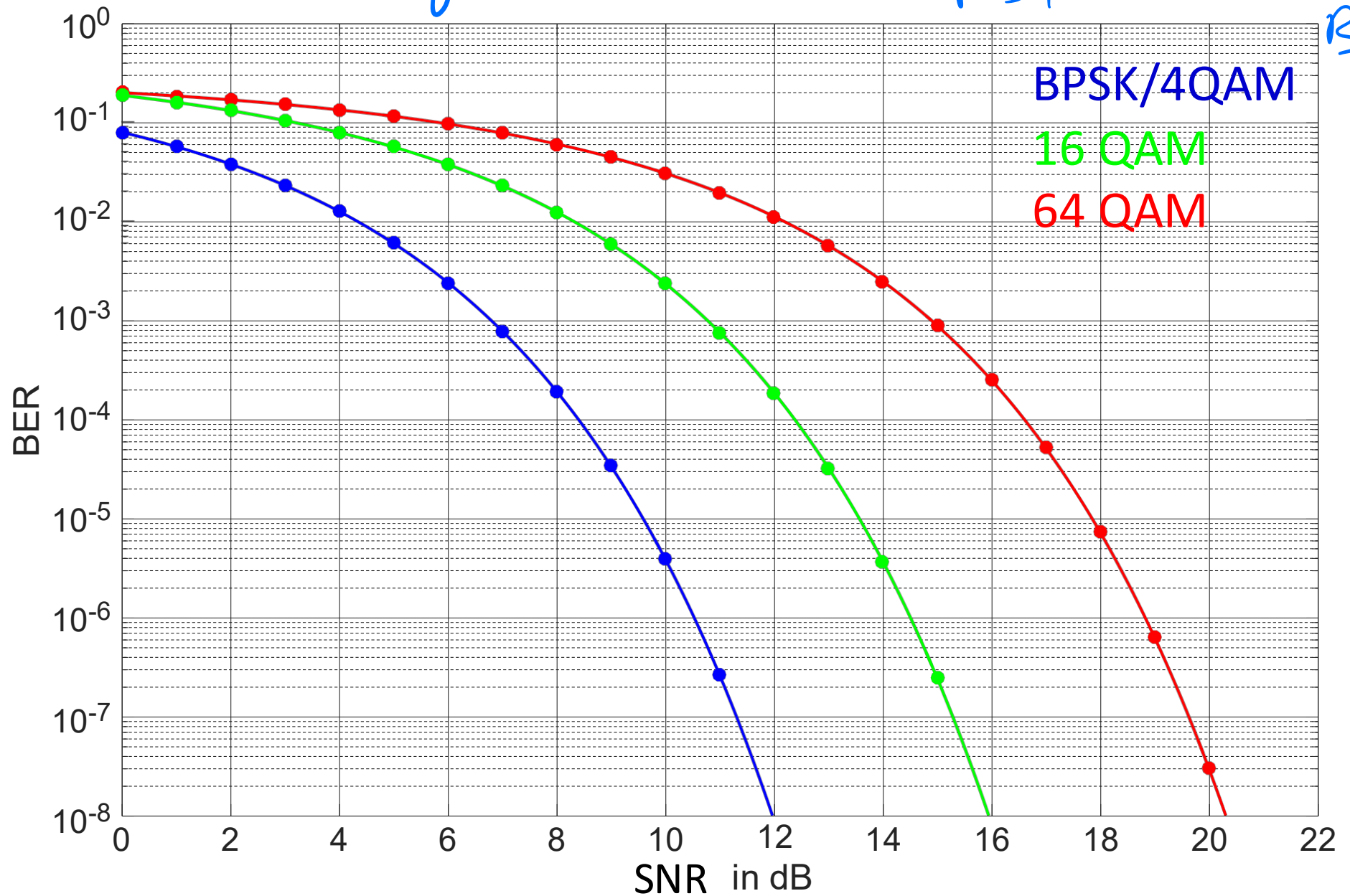
$$n = n_I + j n_Q$$

$$n_I \sim \mathcal{N}(0, \sigma^2)$$

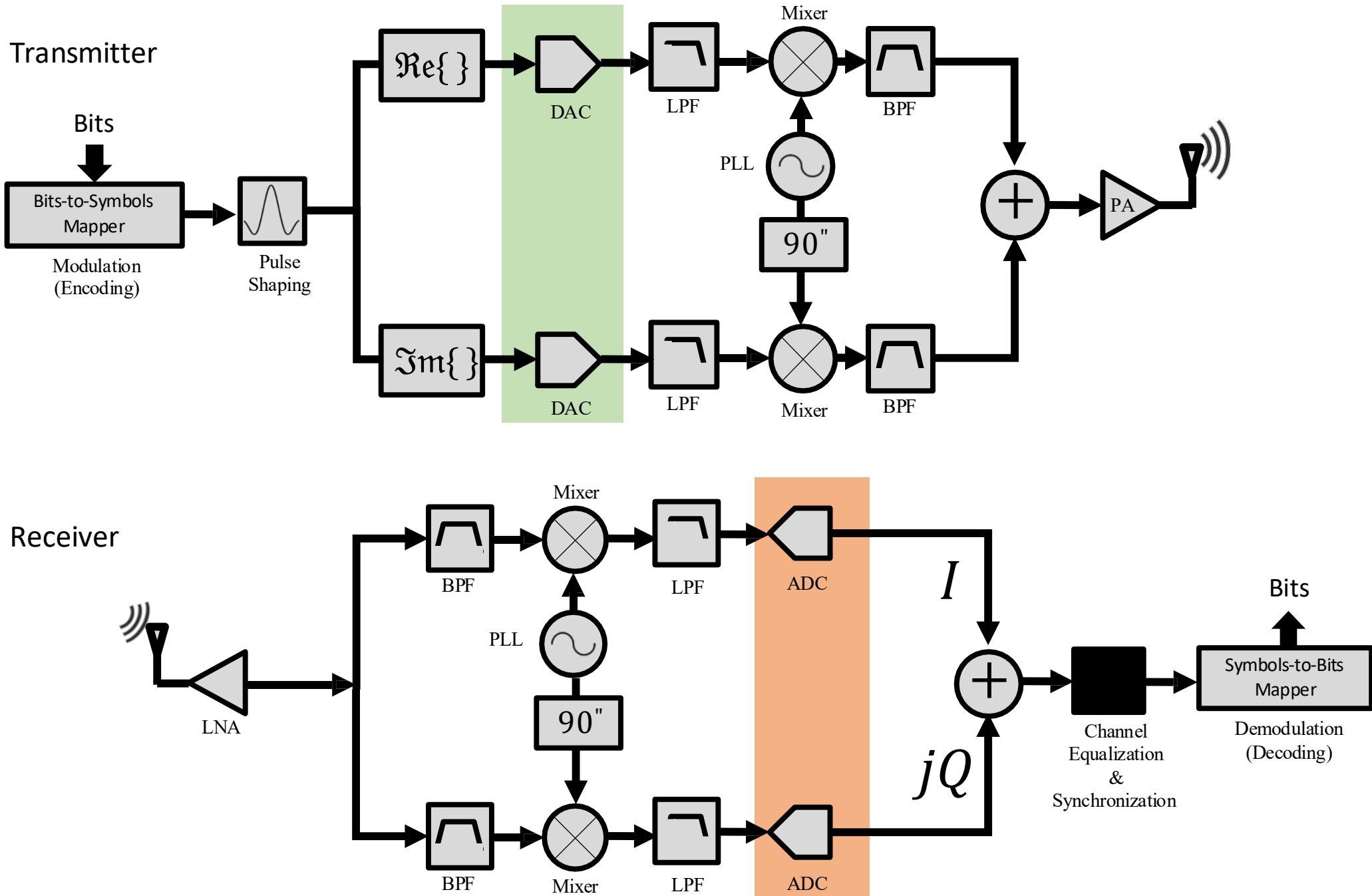
$$n_Q \sim \mathcal{N}(0, \sigma^2)$$

~~Bit Error Rate (BER)~~

Symbol SER QPSK > SER BPSK



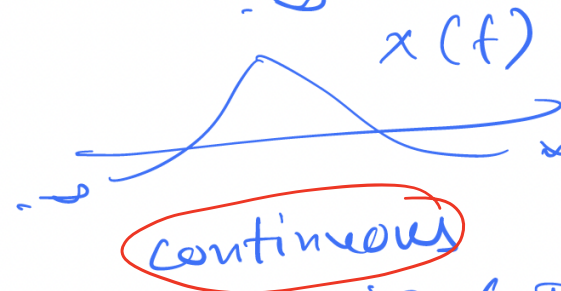
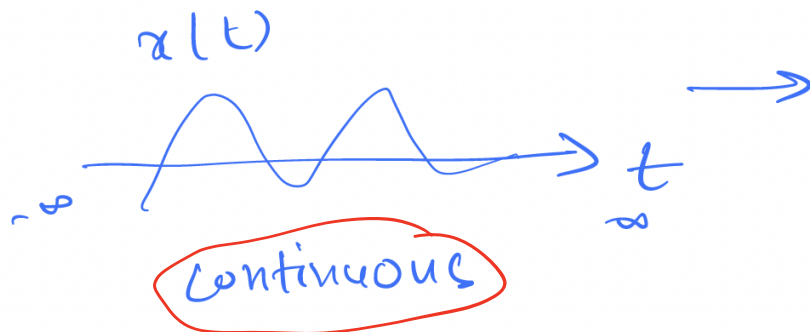
Wireless Communication System



Review: DTFT, DFT

Fourier transform
 $x(t)$ $\rightarrow$

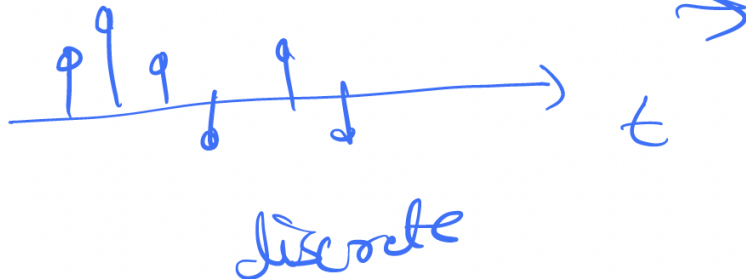
$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$



DTFT

$t: nTs$

$$X(f) = \sum x(nTs) e^{-j2\pi f nTs}$$

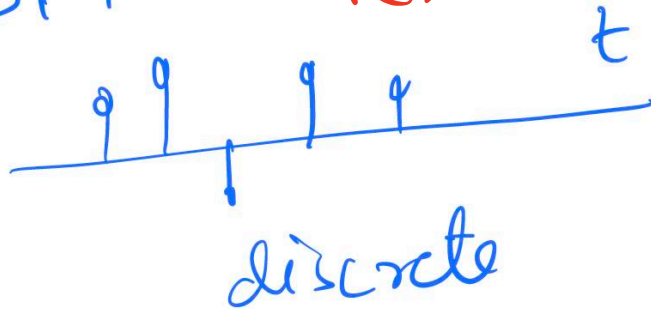


Review: DTFT, DFT

Review: DTFT, DFT

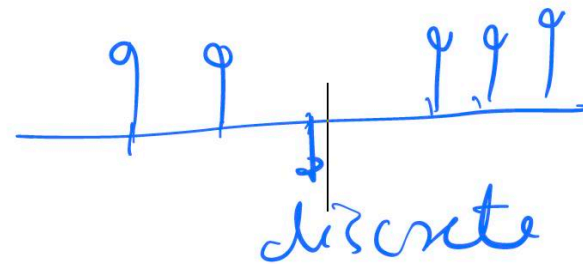
DFT

Periodic

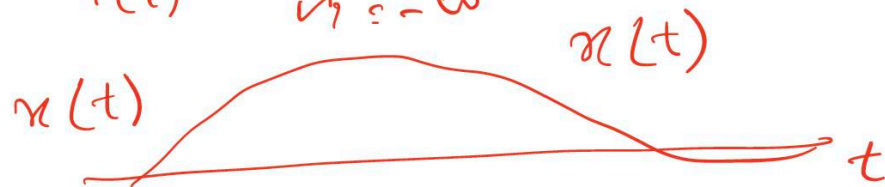
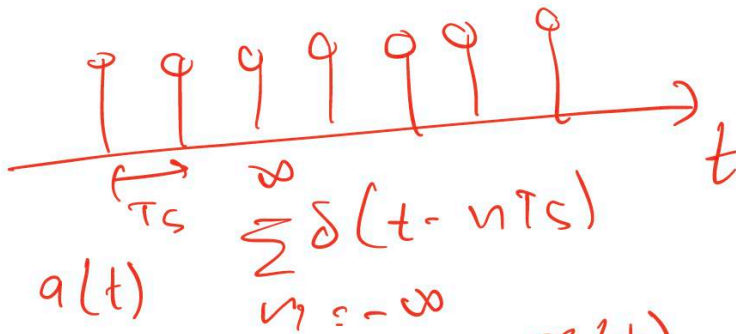


f

Periodic



Pulse train



$$x_\delta(t) = a(t) \times x(t)$$

$$= x(t)$$

$$\times \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

FFT

$$O(N \log N)$$



OFDM

Ideal Sampling

Ideal sampling

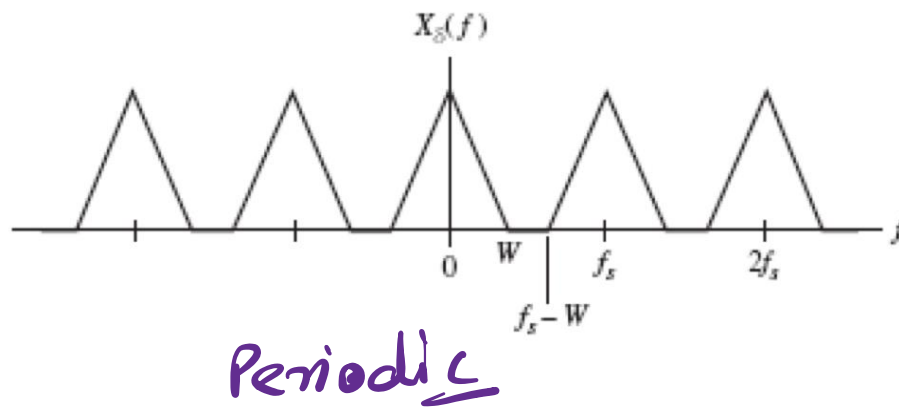
Sampling using "delta" functions

$$x_\delta(t) = \sum_{k=-\infty}^{\infty} x(kT_s) \delta(t - kT_s)$$

$$= x(t) \sum_{k=-\infty}^{\infty} \delta(t - kT_s)$$

Spectrum

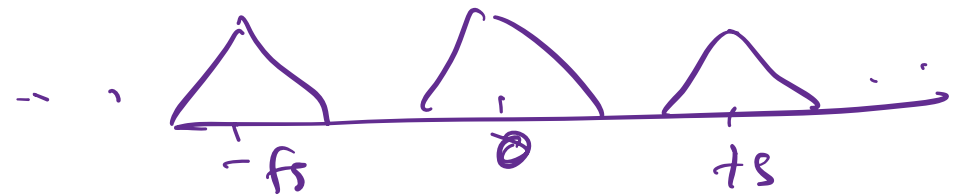
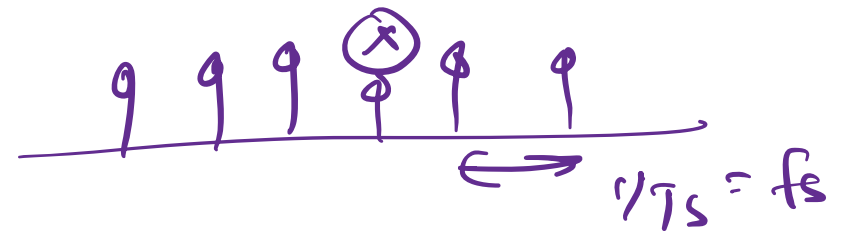
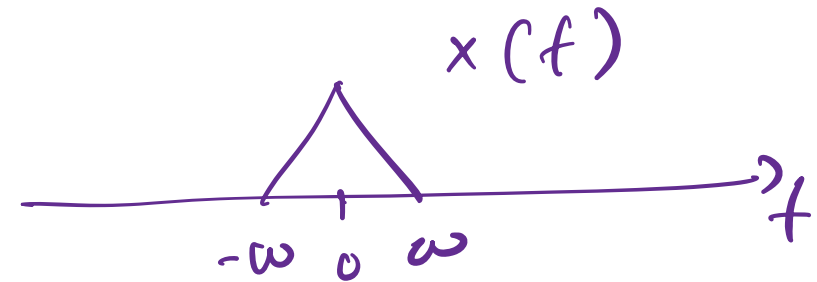
$$X_\delta(f) = f_s \sum_{n=-\infty}^{\infty} X(f - nf_s)$$



Handwritten notes:

$x(t)$

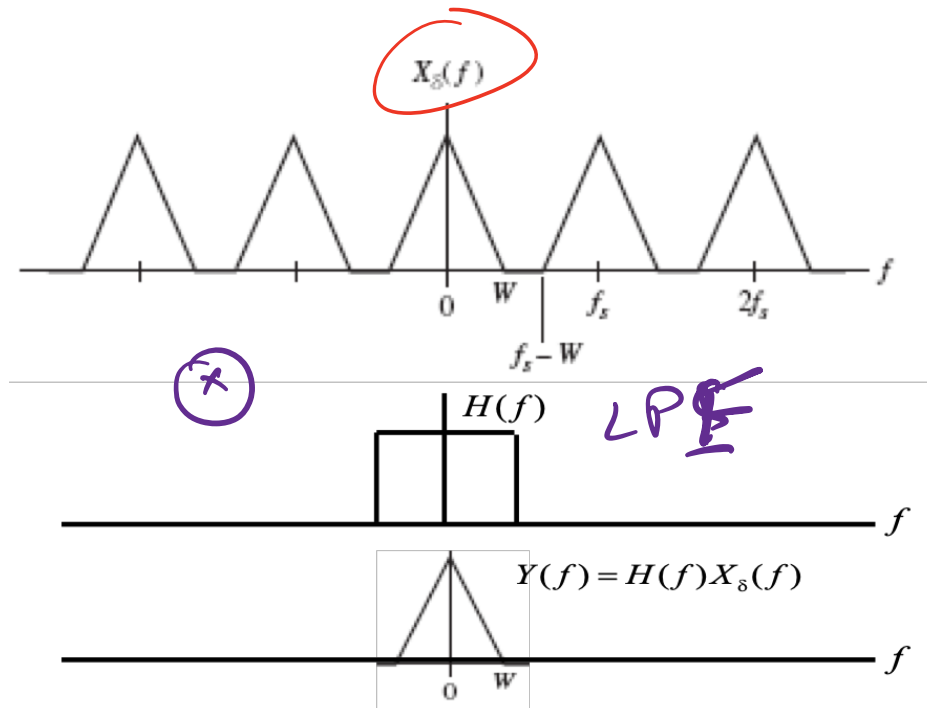
$\sum_n \delta(t - nT_s)$



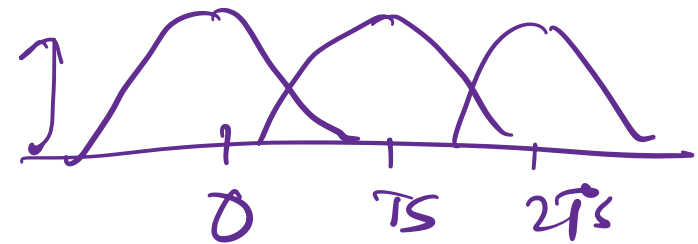
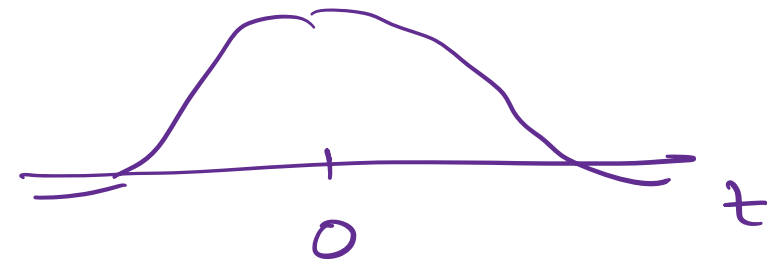
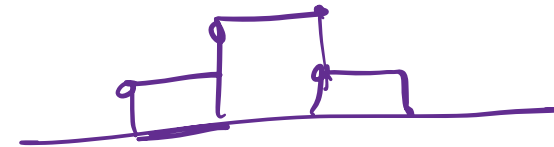
Ideal reconstruction (1/3)

Use an LPF

With cutoff between W and $f_s - W$



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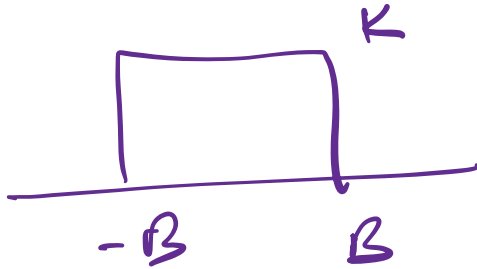


Ideal reconstruction (2/3)

Ideal LPF

$$\underline{H(f)} = K \Pi \left(\frac{f}{2B} \right) e^{-j2\pi f t_d}$$

Reconstructed signal is like original signal, but delayed by t_d , gain K , bandwidth B



Ideal reconstruction (3/3)

Reconstruction by interpolation (time domain)

Reconstruct from samples, but with ideal interpolation

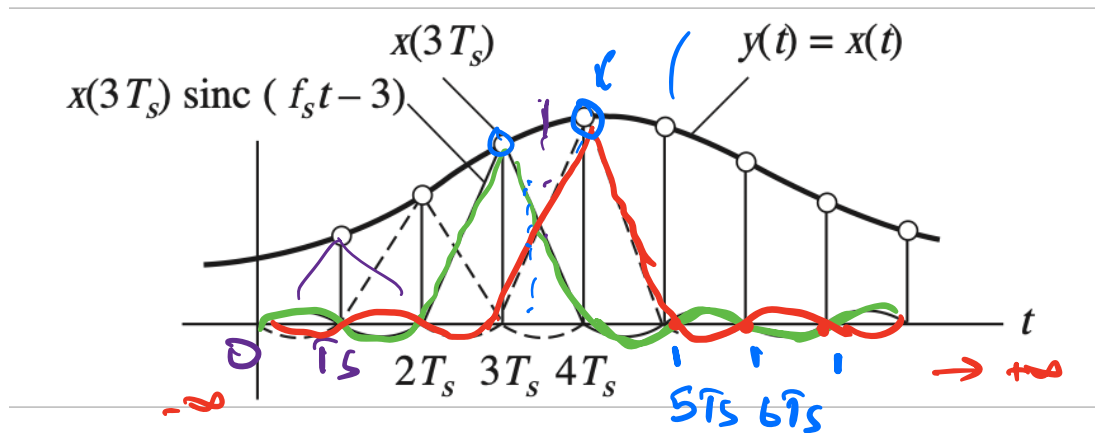
$$h(t) = \mathcal{F}^{-1}[H(f)] = \underline{2BK \operatorname{sinc} 2B(t - t_d)}$$

$$\Rightarrow y(t) = \underbrace{h(t)}_{\text{(Unimplementable in practice)}} * \underbrace{x(kT_s)}_{\text{continuous}} = 2BK \sum_{k=-\infty}^{\infty} \underbrace{x(kT_s)}_{\text{continuous}} \underbrace{\operatorname{sinc} 2B(t - t_d - kT_s)}_{\text{continuous}}$$

↑
continuous

↑
continuous

$x(4T_s)$ $x(5T_s)$



Practical sampling & reconstruction

Real samplers

- Have finite duration pulses (not delta functions)

Real reconstruction filters

- Not ideal (or distortion-less filters)
 - Must be causal (in most cases)
- 

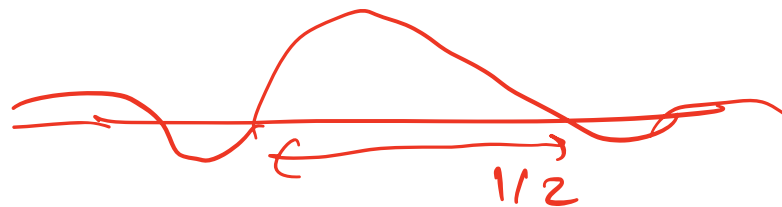
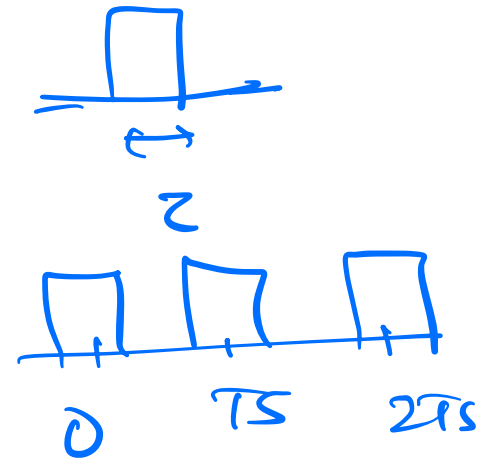
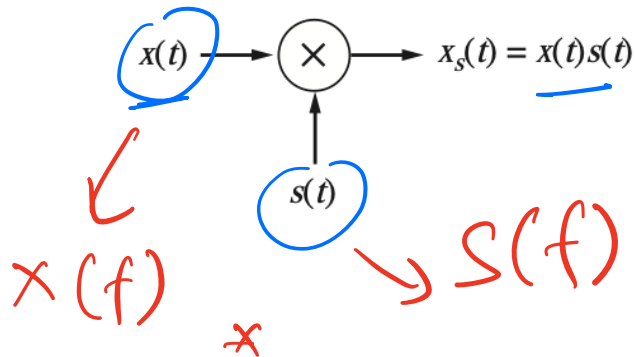
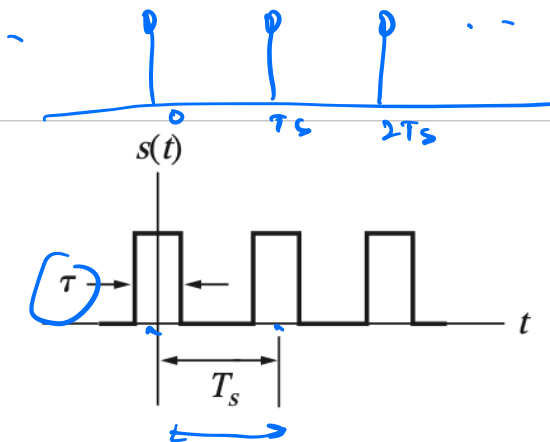
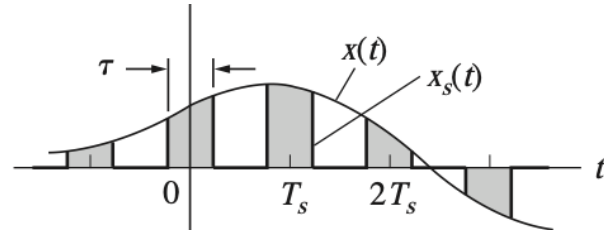
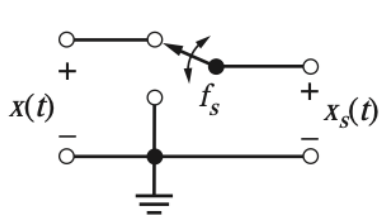
Real signals

- Are time-limited, and therefore cannot be band-limited
- $\Rightarrow$ Signal *distortion* and *aliasing* cannot be entirely avoided when using sampling and reconstruction

Non-ideal sampling (chopper sampling)

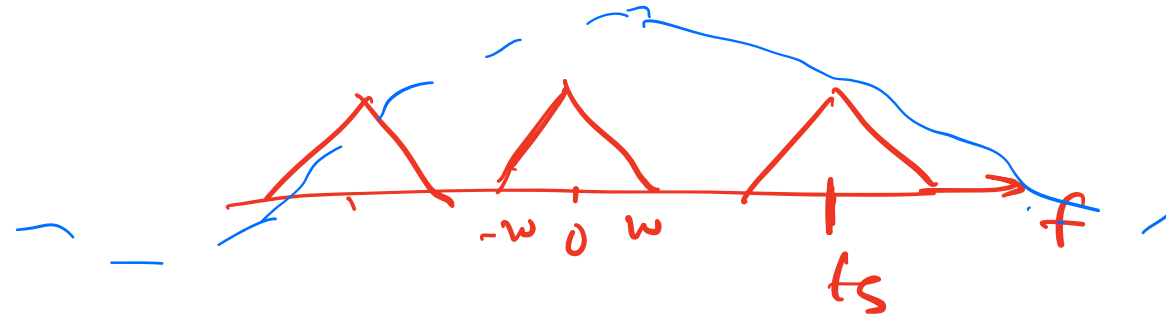
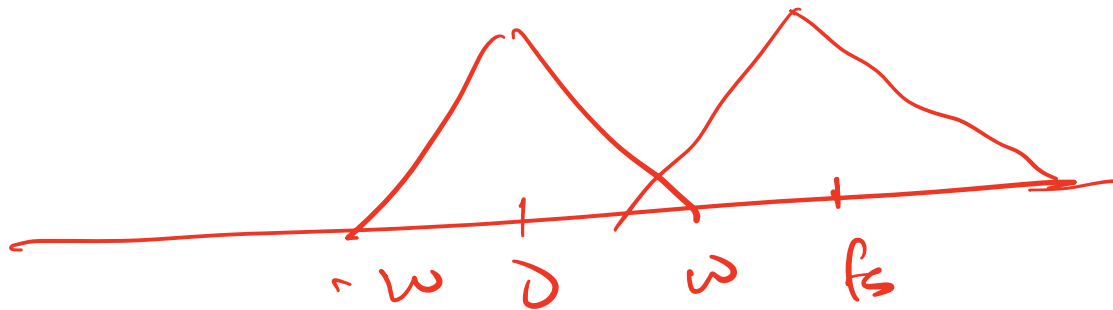
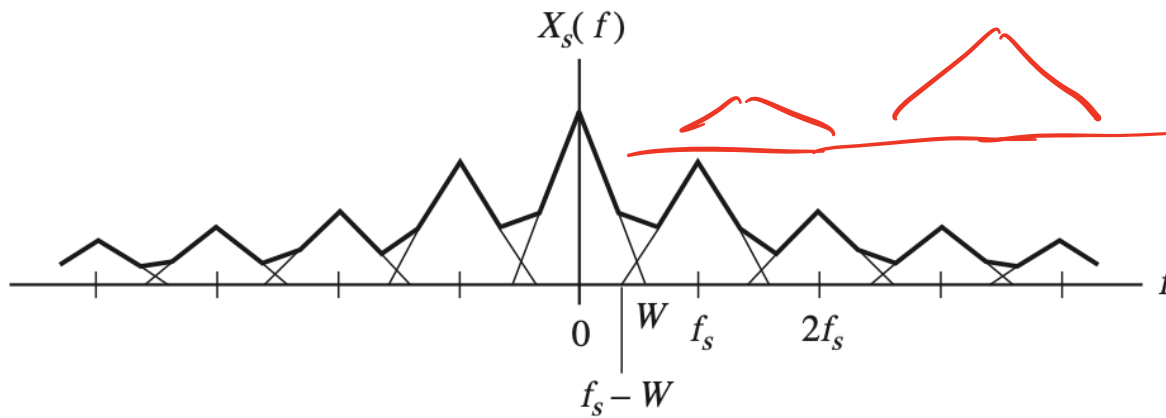
Pulse sampling or chopper sampling

$x_s(t)$: A sampled version of $x(t)$



Aliasing

Under-sampled message



$$f_s > 2w$$

No aliasing

$$f_s < 2w$$

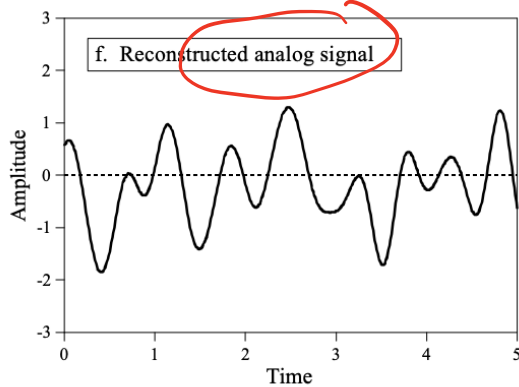
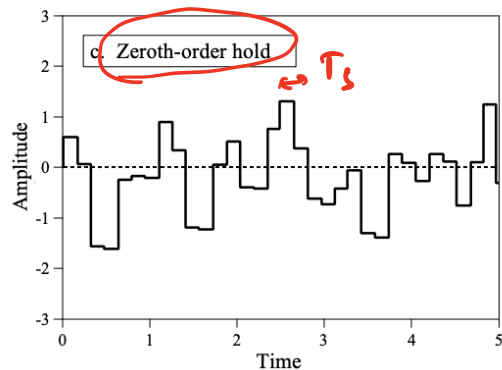
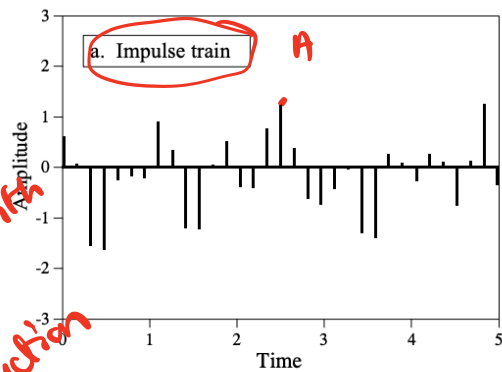
aliasing

Nyquist

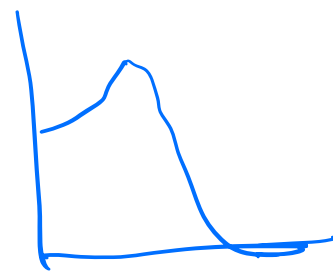
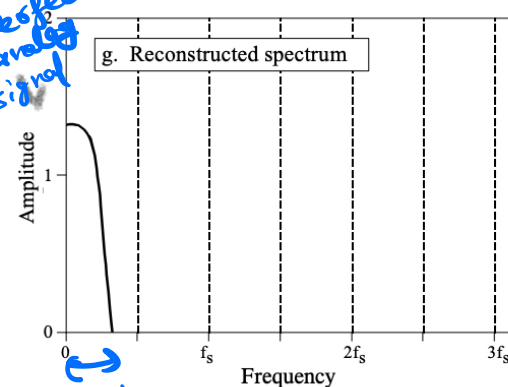
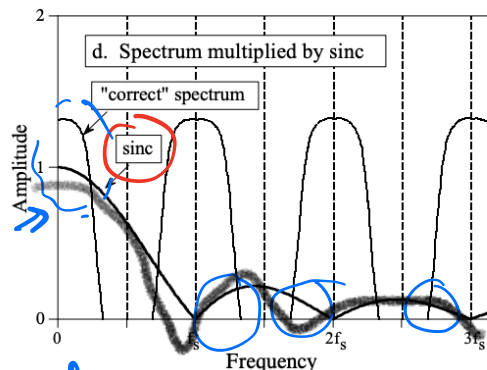
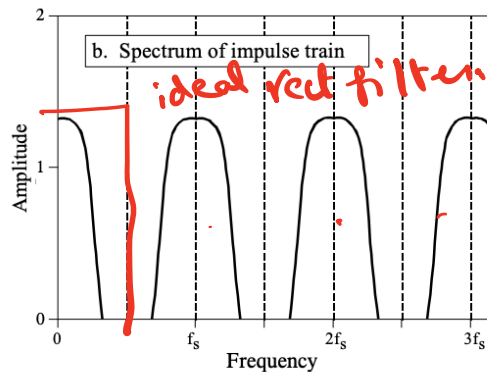
Digital
to
Analog

DAC Design and reconstruction

Time Domain



Frequency Domain



filter design
low-pass.

conv. with
rect
function

filter

perfect
analog
signal

DAC Design and reconstruction

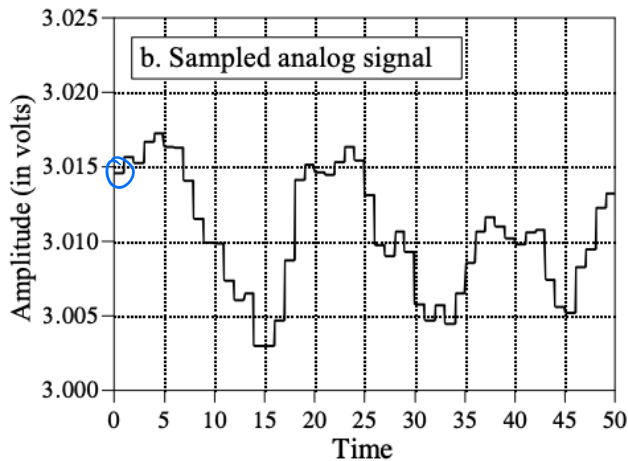
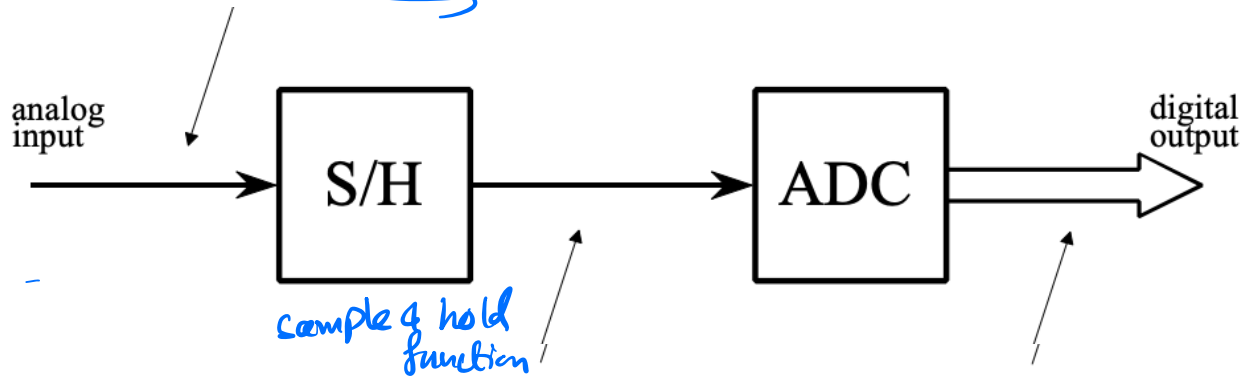
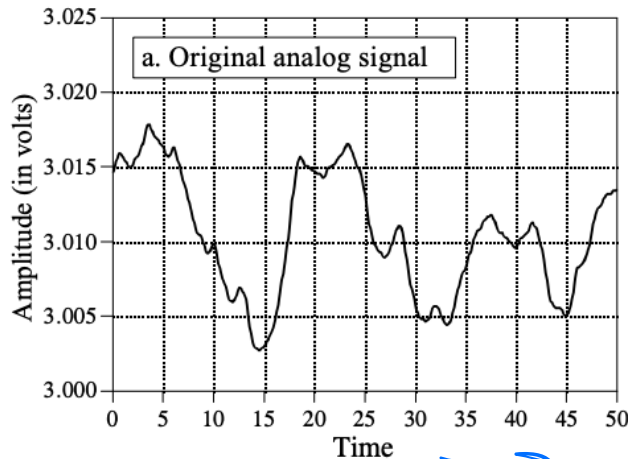
ADC Design with Quantization

Analog
to
Digital

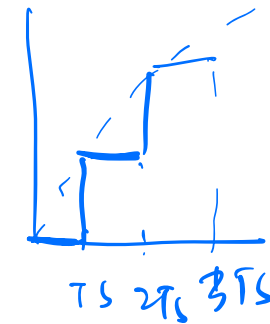
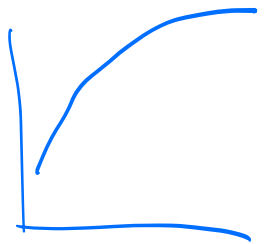
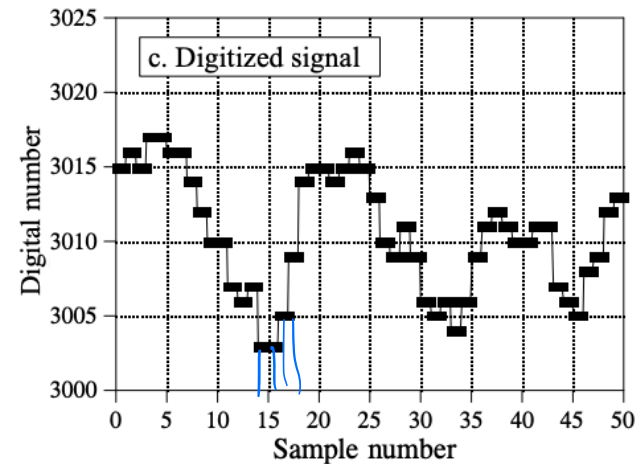
$$f_s = \text{sampling rate}$$

$$= \frac{1}{T_s}$$

$$T_s = \frac{1}{f_s}$$

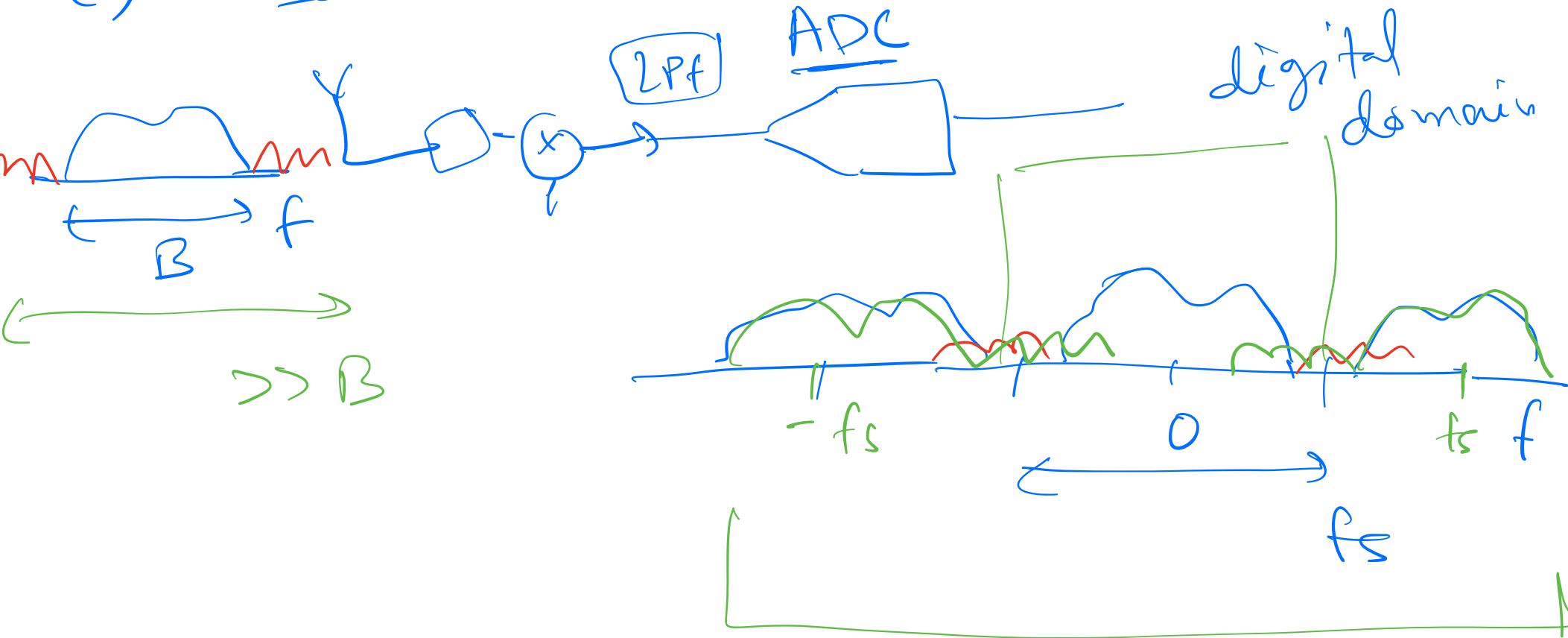


quantized

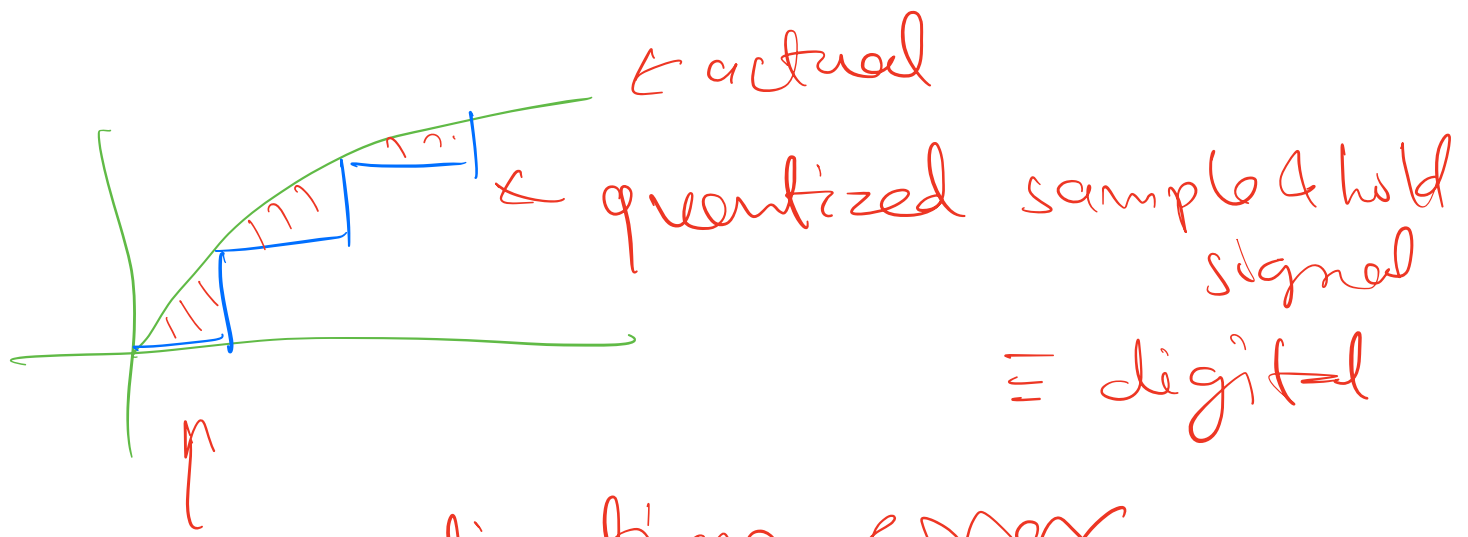


ADC Design with Quantization

(1) Nyquist - sampling



(2) Quantization 12 bit - 16 bit -
2 bit - noise

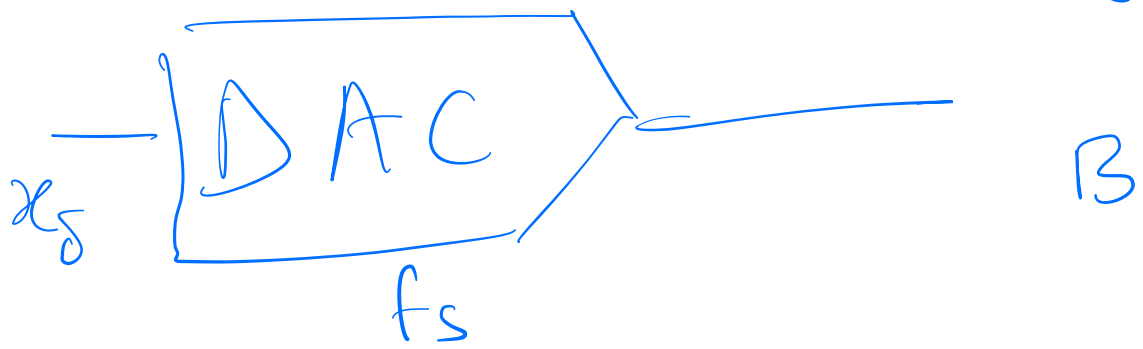


↑ bit ADC $\Rightarrow$ ↓ low quantiz. error.

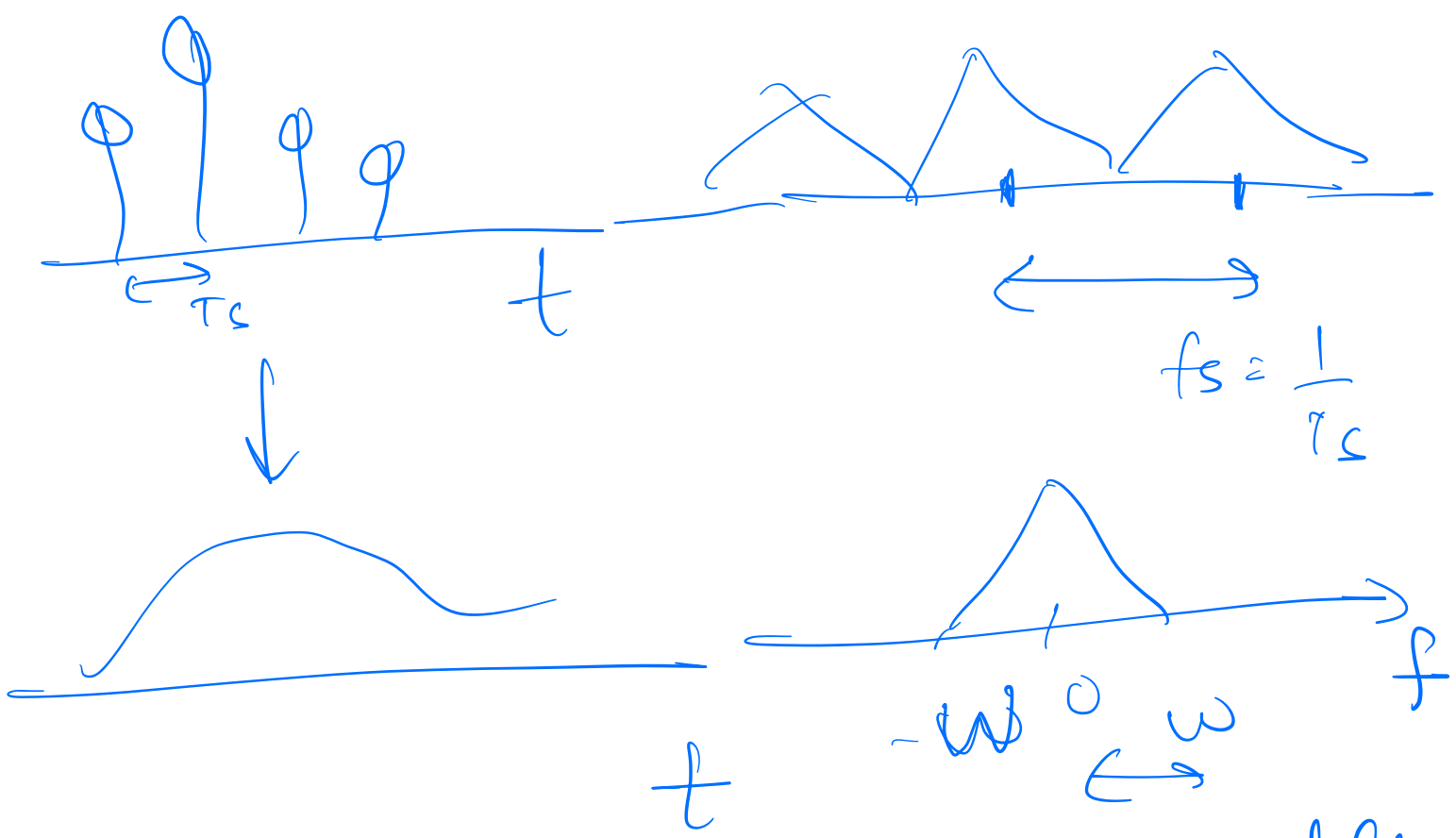
$\Rightarrow$ ↑ SAR.

samples

analog signal $x(t)$



Q what is signal bandwidth.

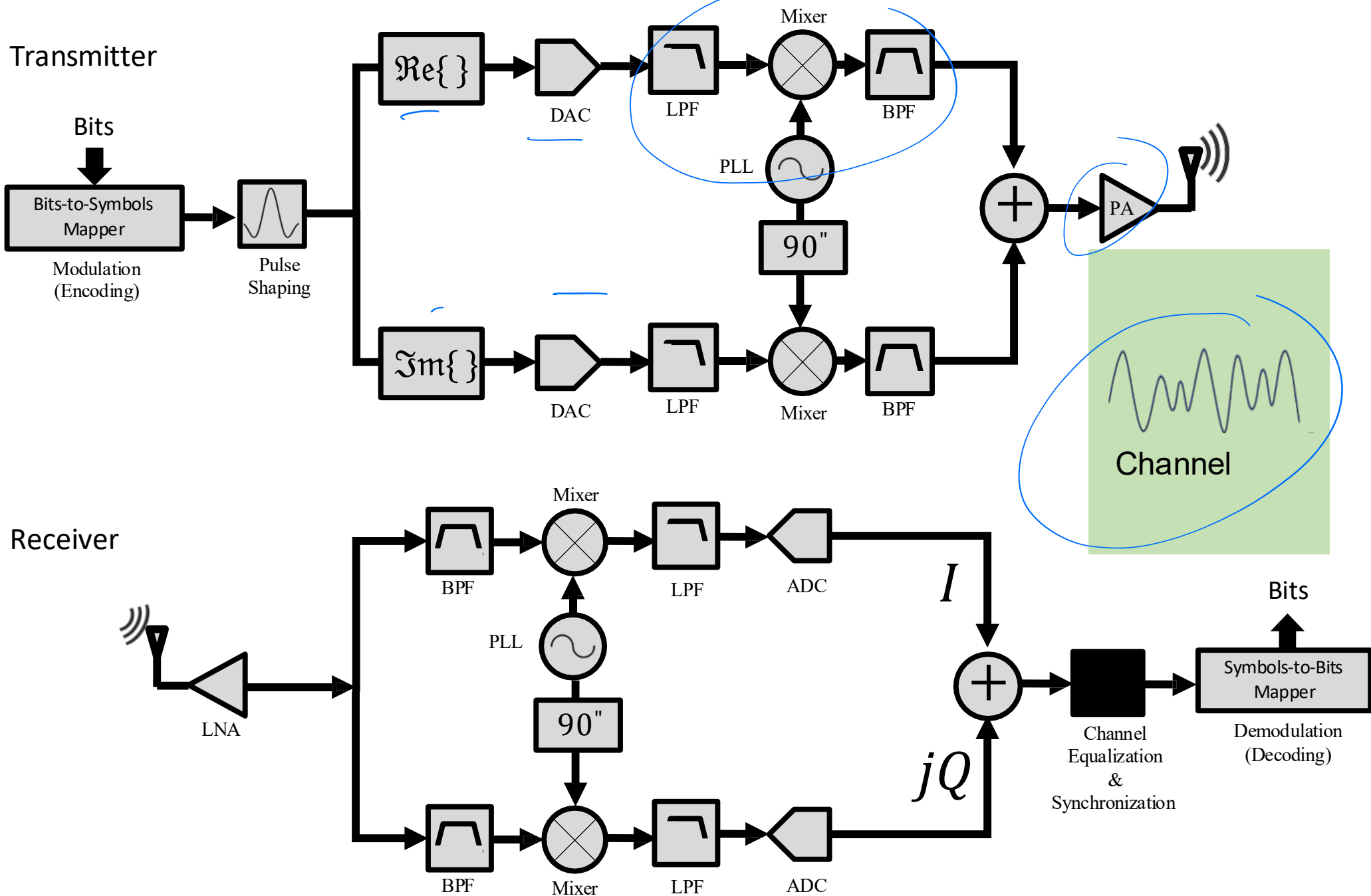


$$f_s = \frac{1}{T_s}$$

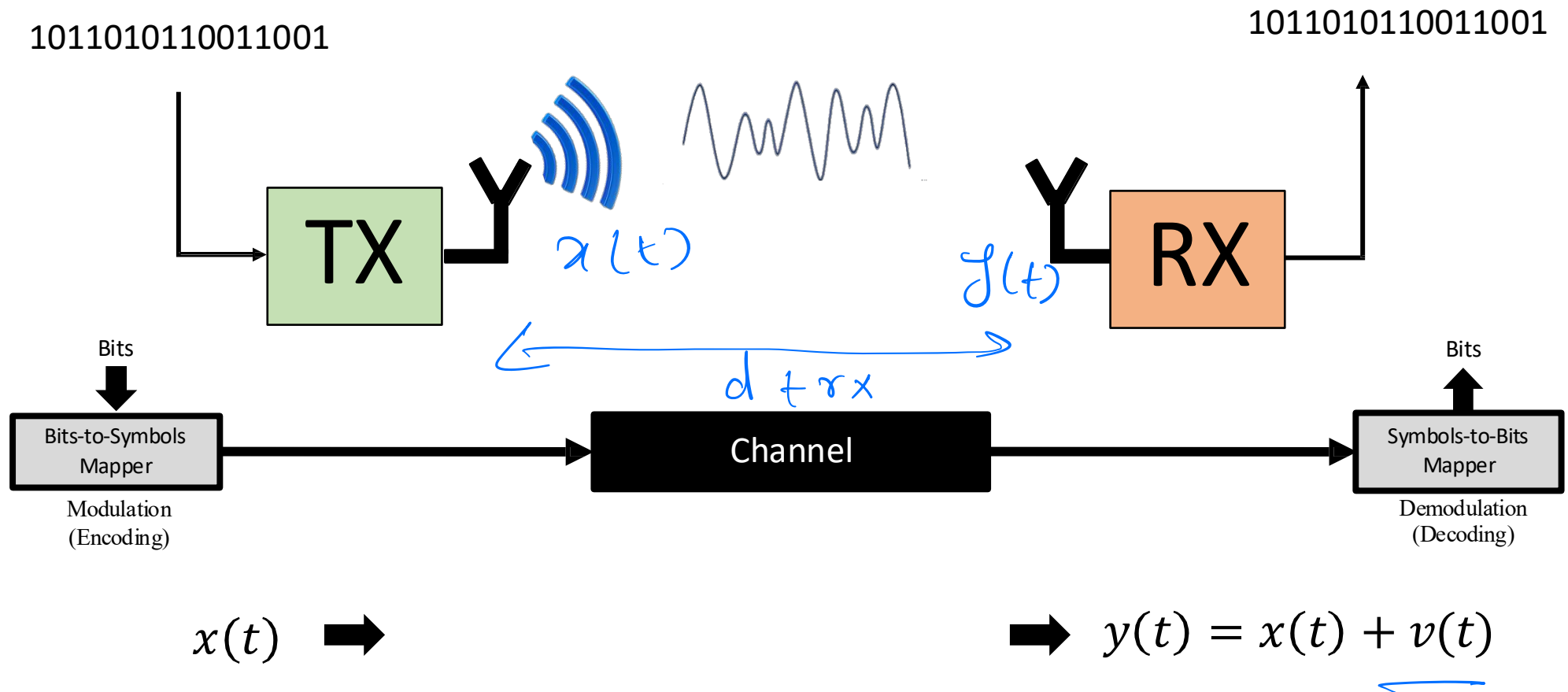
$$W = \frac{f_s}{2}$$

Bandwidth
= W

Wireless Communication System



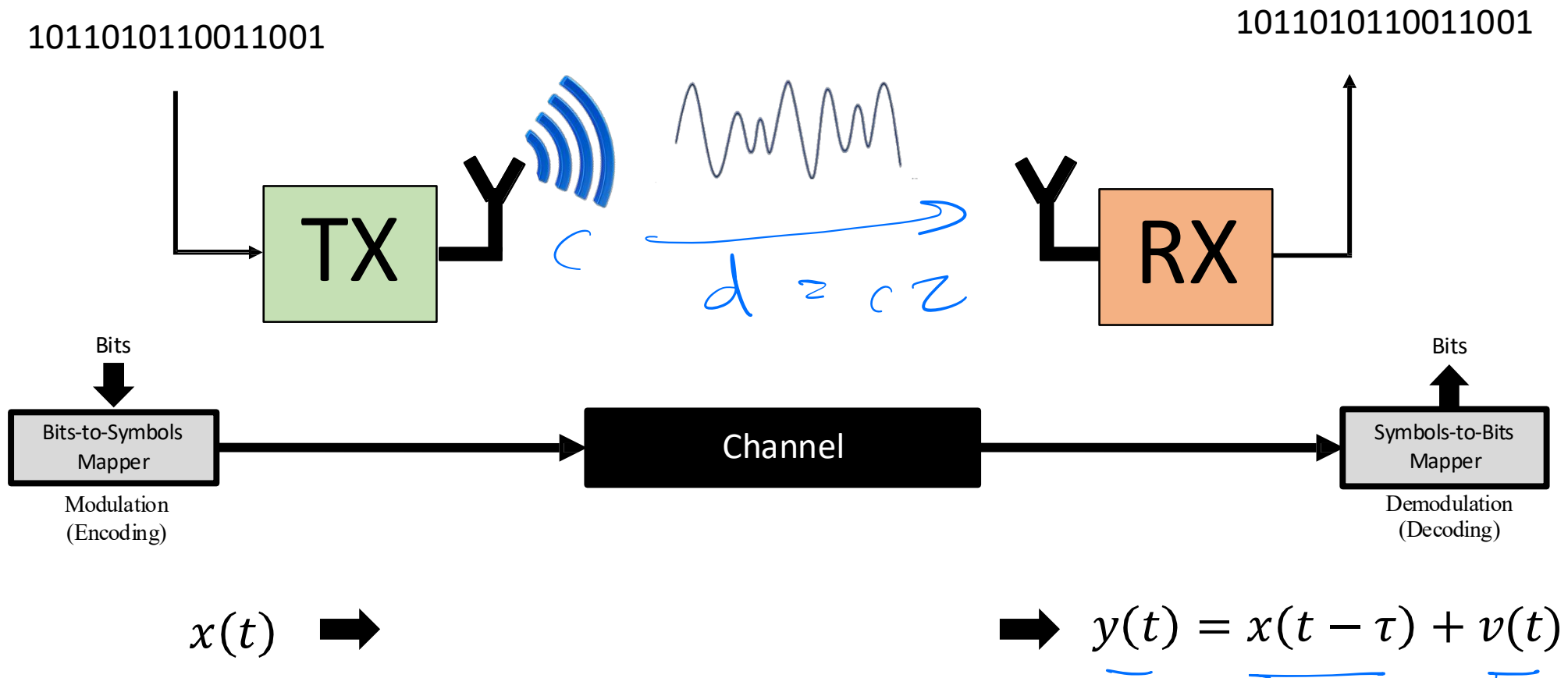
The Channel



Channel adds noise (AWGN)!

$$v(t) \sim N(0, \sigma^2)$$

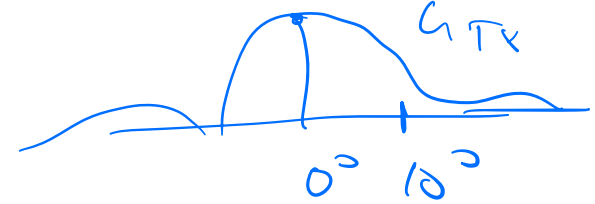
The Channel



Channel delays the signal!

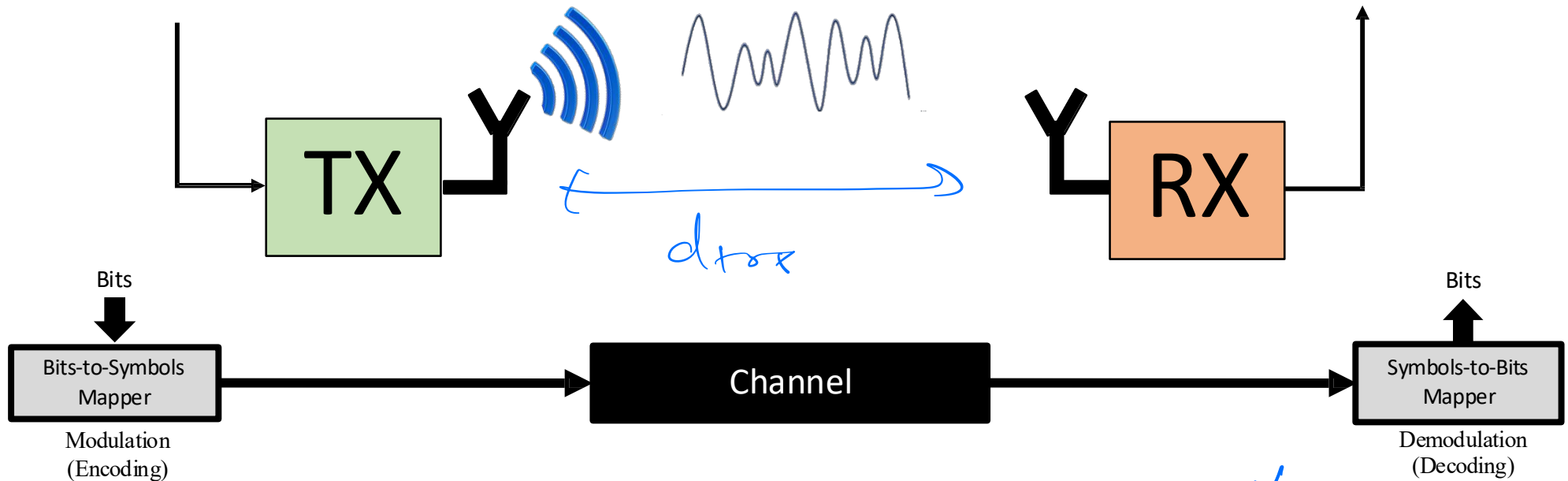
$$\tau = \frac{d}{c}$$

The Channel



1011010110011001

1011010110011001



$$\lambda = \frac{c}{f}$$

$$f = \frac{c}{\lambda}$$

$x(t) \rightarrow$

$$\rightarrow y(t) = \mathbf{h} x(t - \tau) + v(t)$$

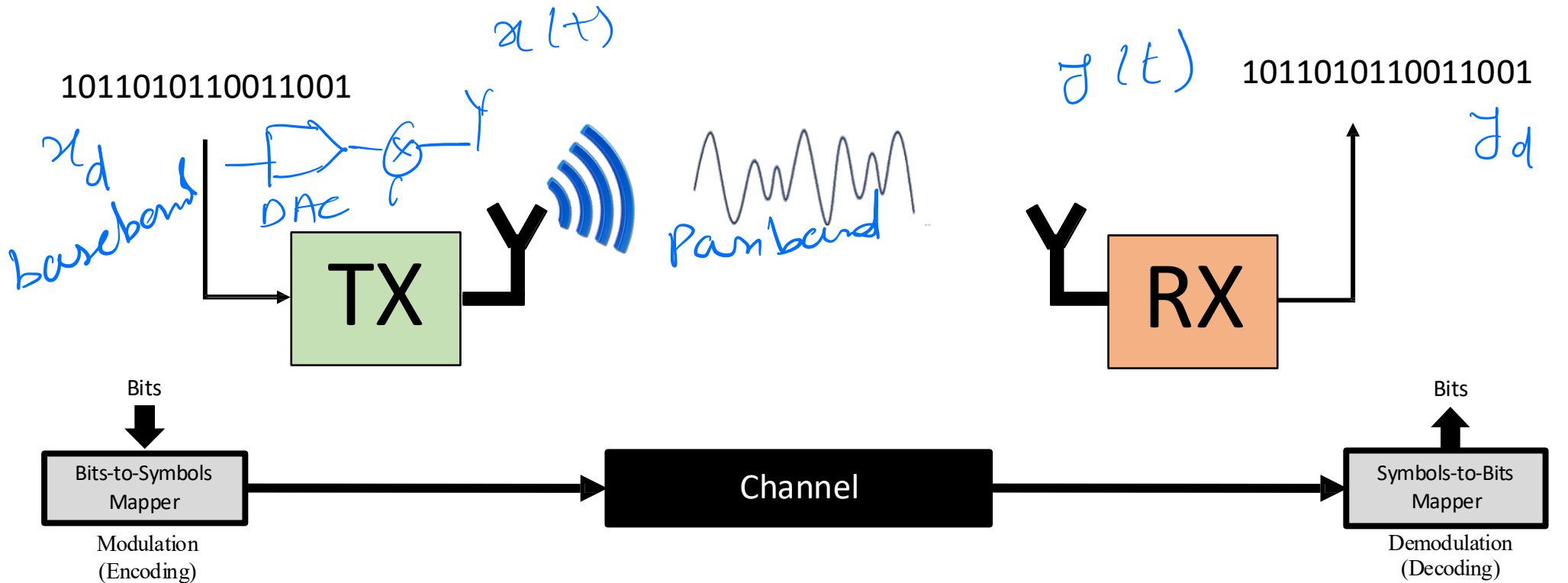
Channel attenuates the signal (Pathloss)

$$P_{RX} = G_{TX} G_{RX} \frac{\lambda^2}{(4\pi d)^2} P_{TX}$$

Handwritten annotations: G_{TX} is labeled 'TX antenna gain', G_{RX} is labeled 'Rx', P_{TX} is labeled 'Tx power', and d is labeled 'd_tx rx'.

$$\Rightarrow |h| \propto \frac{\lambda}{d}$$

The Channel



$x(t) \rightarrow$

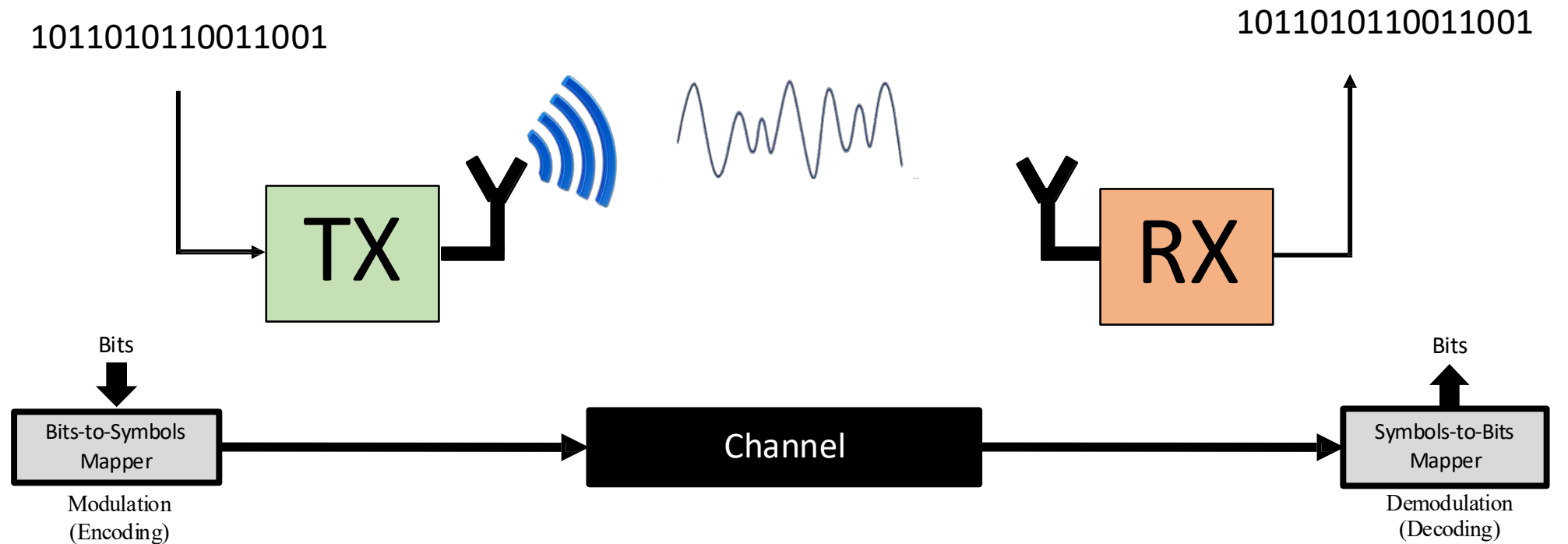
$$\rightarrow \underline{y(t) = \underline{h} x(t - \tau) + v(t)}$$

Channel rotates the signal (Adds Phase)

$$h \propto \frac{\lambda}{d} e^{j\phi}$$

$$e^{j\phi}$$

The Channel



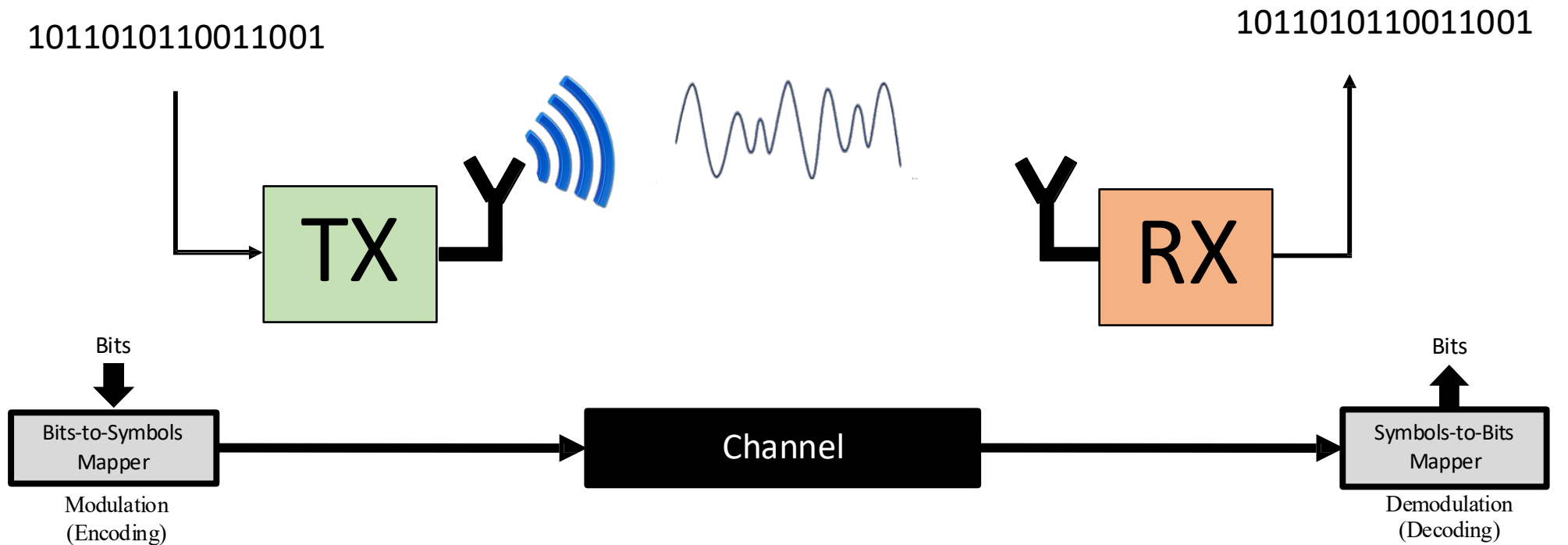
$$x(t) \Rightarrow$$

$$\Rightarrow y(t) = \mathbf{h} x(t - \tau) + v(t)$$

$$x(t) \times e^{-j2\pi f_c t} \Rightarrow |h| x(t - \tau) e^{-j2\pi f_c (t - \tau)} \Rightarrow x(t) e^{j2\pi f_c t} \Rightarrow |h| x(t - \tau) e^{j2\pi f_c t}$$

$$\mathbf{h} \propto \frac{\lambda}{d} e^{j\phi} \rightarrow \phi = 2\pi f_c \tau = 2\pi \frac{c}{\lambda} \frac{d}{c} = 2\pi \frac{d}{\lambda} \rightarrow \mathbf{h} \propto \frac{\lambda}{d} e^{j2\pi d/\lambda}$$

The Channel



$$x(t) \Rightarrow$$

$$\Rightarrow y(t) = \mathbf{h} x(t - \tau) + v(t)$$

Channel:

- Adds Noise
- Delays the Signal
- Attenuates the Signal
- Rotates the Phase of the Signal

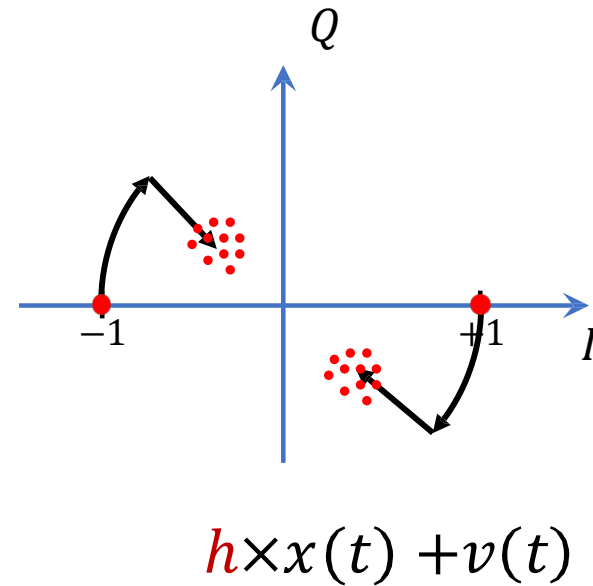
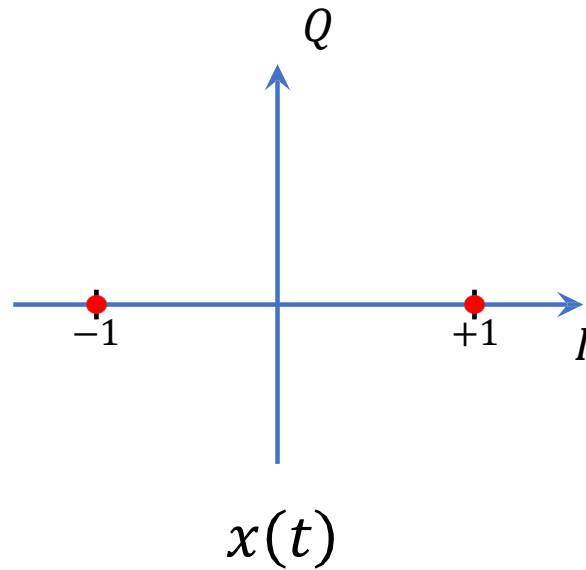
$$\mathbf{h} \propto \frac{\lambda}{d} e^{j2\pi d/\lambda}$$

The Channel

Consider BPSK Modulation.

$$0 \rightarrow -1$$

$$1 \rightarrow +1$$

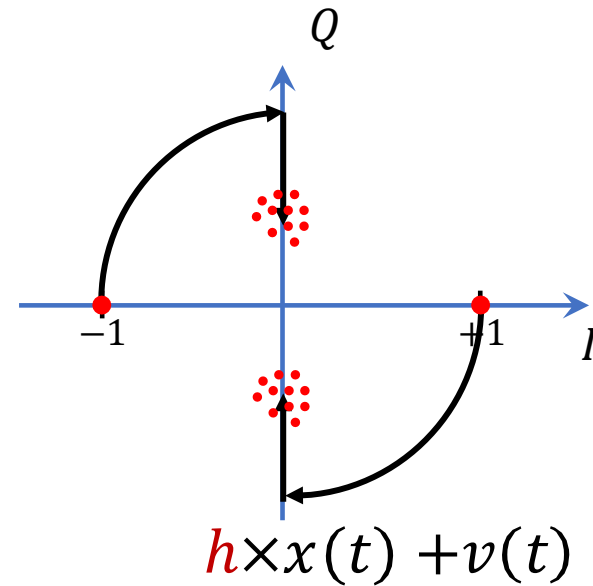
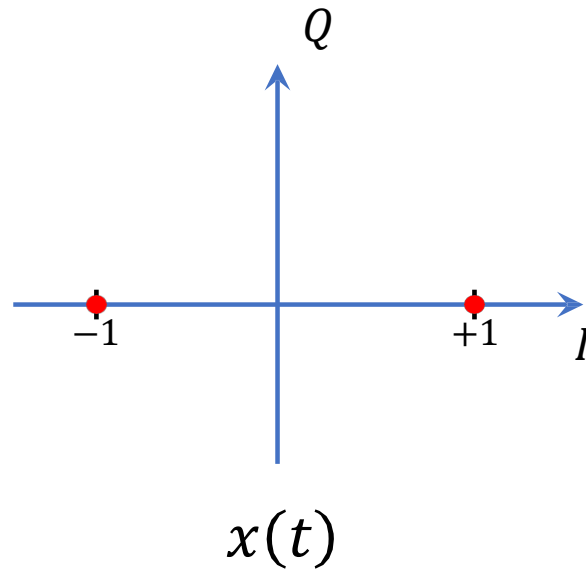


The Channel

Consider BPSK Modulation.

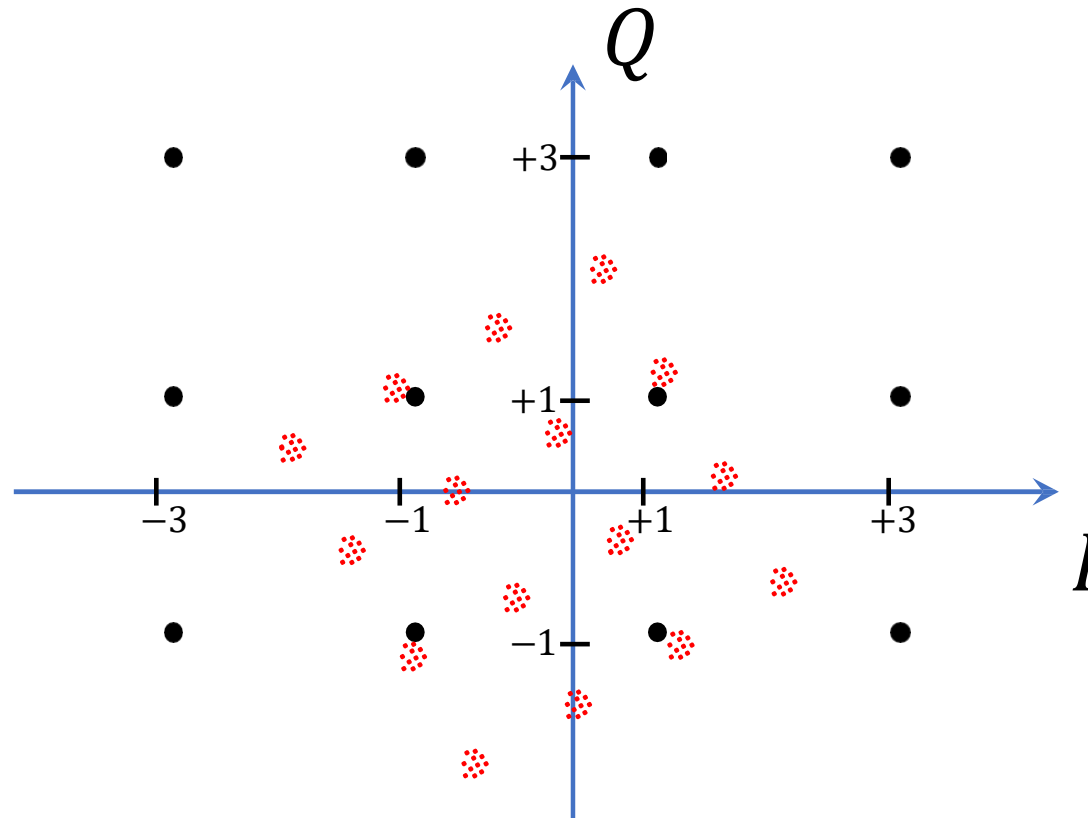
$$0 \rightarrow -1$$

$$1 \rightarrow +1$$



The Channel

Consider QAM Modulation



Demodulating correctly requires COHERENCE!
i.e., Need to estimate & correct for the channel h



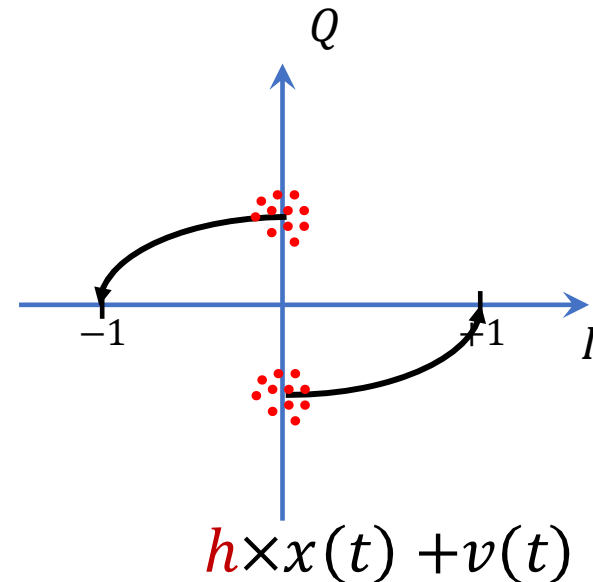
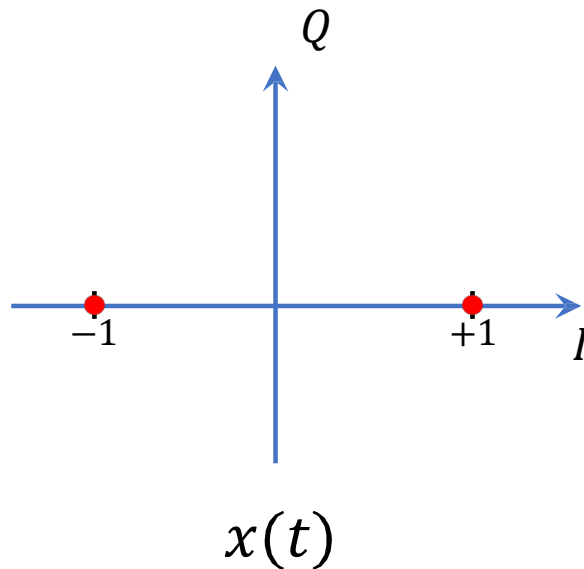
Channel Estimation & Correction

Channel Estimation & Correction

Consider BPSK Modulation.

$$0 \rightarrow -1$$

$$1 \rightarrow +1$$



Send Training Sequence (Preamble Bits): Known Bits

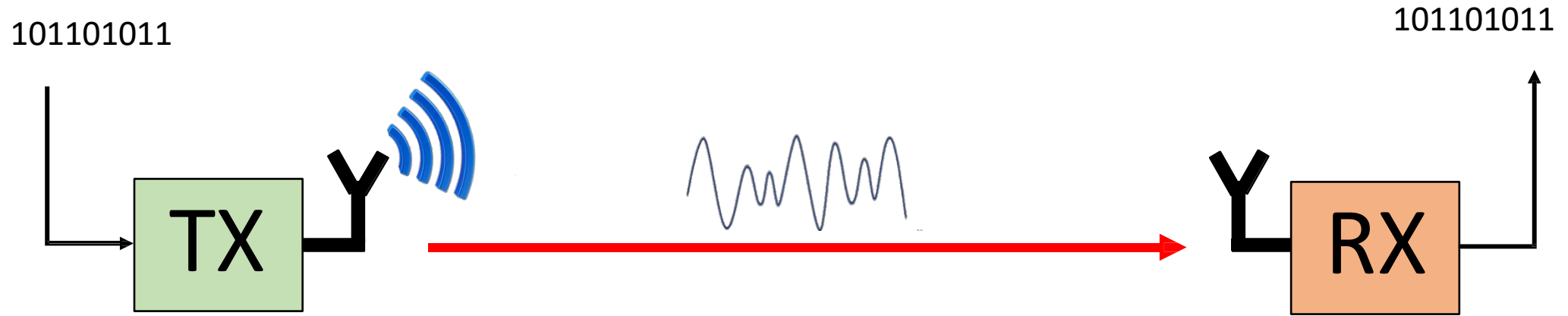
$$\begin{aligned} x(0) = 1 & \longrightarrow y(0) = h + v(0) \\ x(1) = 1 & \longrightarrow y(1) = h + v(1) \\ x(2) = -1 & \longrightarrow y(2) = -h + v(2) \end{aligned}$$

•
•
•

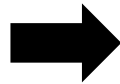
$$\text{Estimate channel: } \tilde{h} = \frac{1}{K} \sum_{k=1}^K \frac{y(k)}{x(k)}$$

$$\text{Correct channel: } \tilde{x}(t) = \frac{y(t)}{\tilde{h}(t)}$$

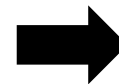
The Channel



$x(t)$



Channel

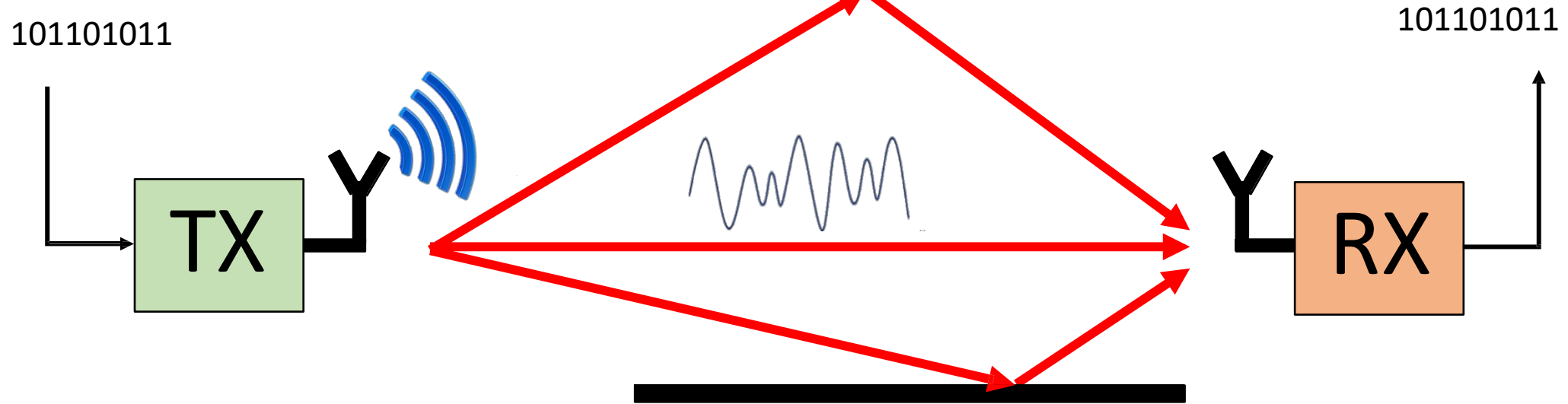


$y(t) = \mathbf{h} x(t - \tau) + v(t)$

$$h \propto \frac{\lambda}{d} e^{j2\pi d/\lambda}$$

Assumes single path!

Multipath Channel



Multipath Propagation: radio signal reflects off objects ground, arriving at destination at slightly different times

$$y(t) = \alpha_1 e^{j\phi_1} x(t - \tau_1) + \alpha_2 e^{j\phi_2} x(t - \tau_2) + \alpha_3 e^{j\phi_3} x(t - \tau_3) \dots$$

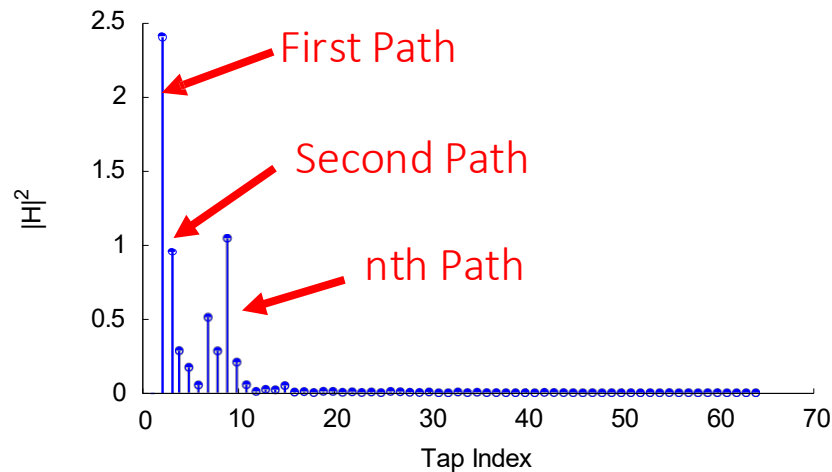
$$y(t) = \sum_k \alpha_k e^{j\phi_k} x(t - \tau_k) = \sum_k h_k(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

$h(t)$ is channel impulse response.

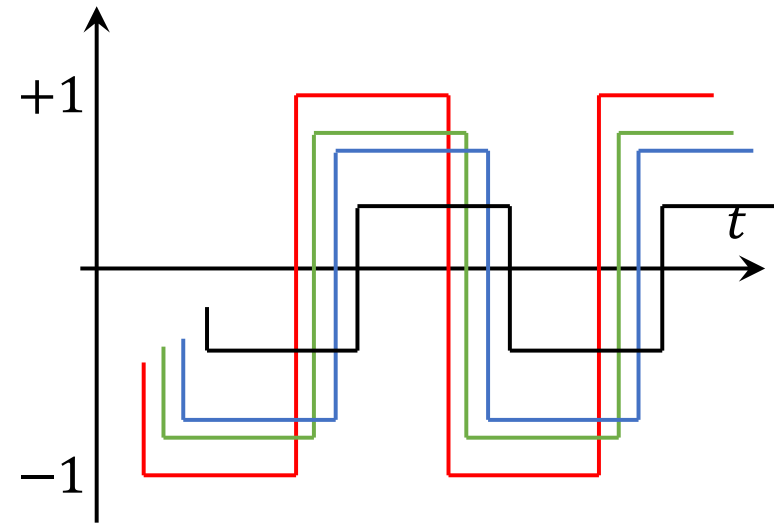
Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$



Multi-tap Channel



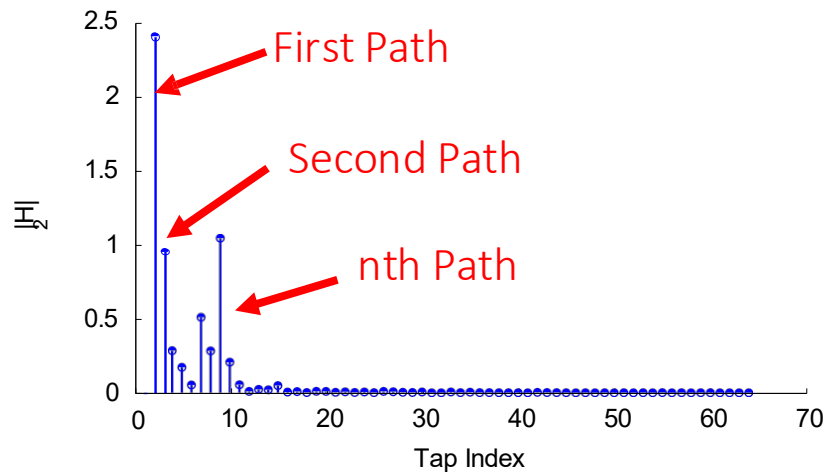
ISI: Inter-Symbol-Interference

Symbols arriving along late paths interfere with following symbols.

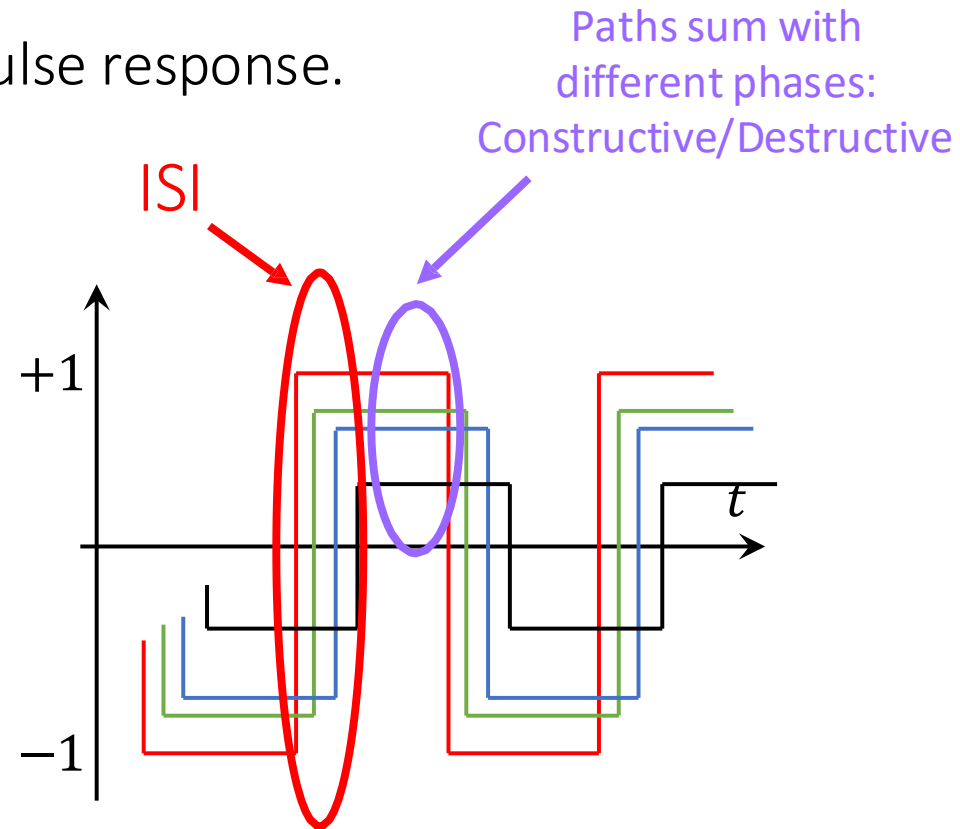
Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$



Multi-tap Channel



ISI: Inter-Symbol-Interference

Symbols arriving along late paths interfere with following symbols.



Channel Fading

Symbols arriving along different paths sum up destructively

Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

Channel Fading: Symbols arriving along different paths sum up destructively

Example 2 paths with distance $d_1 = 1m$, $d_2 = 1.06m$:

$$h = h_1 + h_2 = \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} + \frac{\lambda}{d_2} e^{j2\pi d_2/\lambda}$$

$$= \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} \left(1 + \frac{d_1}{d_2} e^{j2\pi(d_2-d_1)/\lambda} \right) \quad \frac{d_1}{d_2} \approx 1$$

$$\text{if } \frac{d_2 - d_1}{\lambda} \approx \frac{1}{2} \rightarrow h = \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} (1 + e^{j\pi}) = 0 \quad \text{Destructive Interference!}$$

Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

Channel Fading: Symbols arriving along different paths sum up destructively

Example 2 paths with distance $d_1 = 1m, d_2 = 1.06m$:

$$h = h_1 + h_2 = \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} + \frac{\lambda}{d_2} e^{j2\pi d_2/\lambda}$$

@ $f_1 = 2.5GHz$ ($\lambda = 12\text{ cm}$): $h = 0.12 e^{j\frac{2\pi}{3}} + 0.113 e^{j\frac{5\pi}{3}} \approx 0.006$

@ $f_2 = 5GHz$ ($\lambda = 6\text{ cm}$): $h = 0.06 e^{j\frac{5\pi}{3}} + 0.05 e^{j\frac{5\pi}{3}} \approx 0.116$

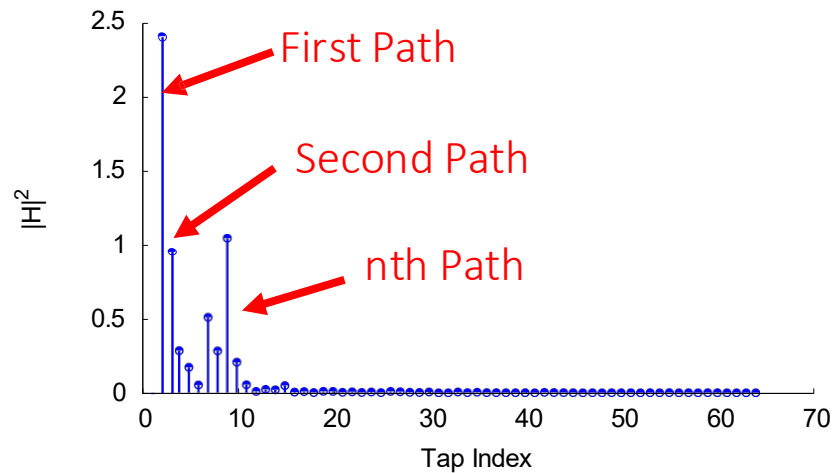
Frequency Selective Fading

17×
(24dB)

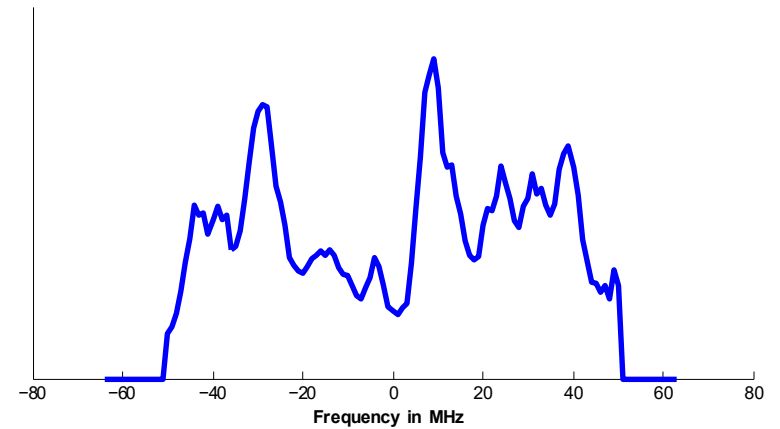
Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



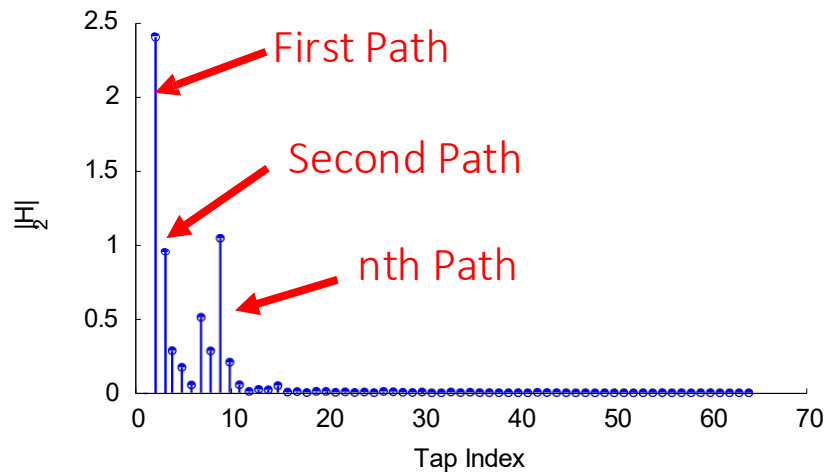
Multi-tap Channel



Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$

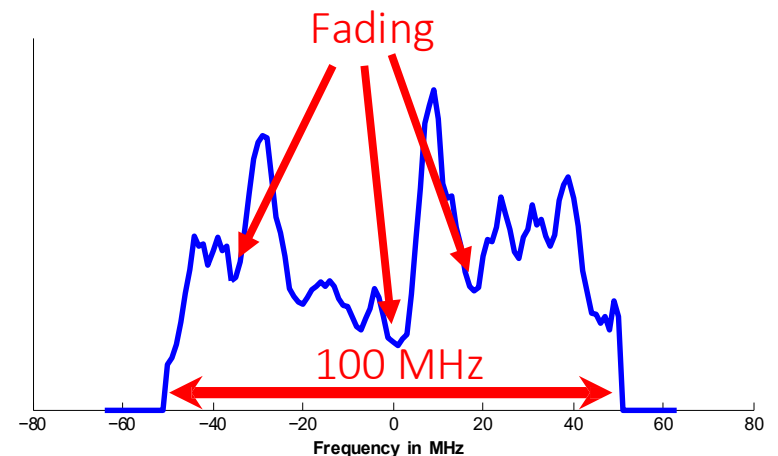


Multi-tap Channel



ISI: Inter-Symbol-Interference

Symbols arriving along late paths interfere with following symbols.



Frequency Selective Fading

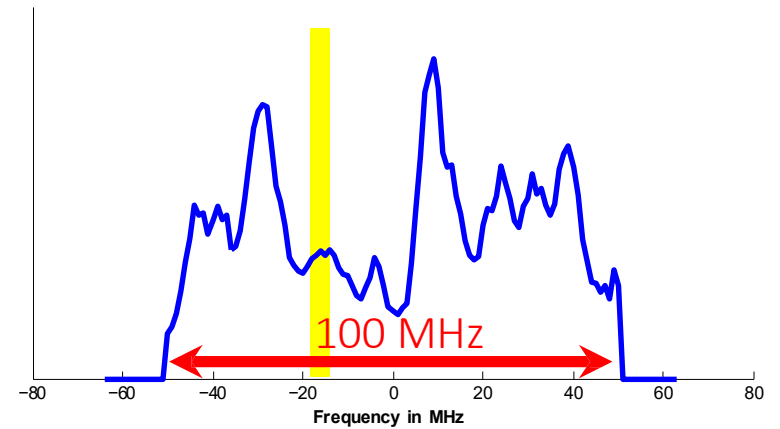
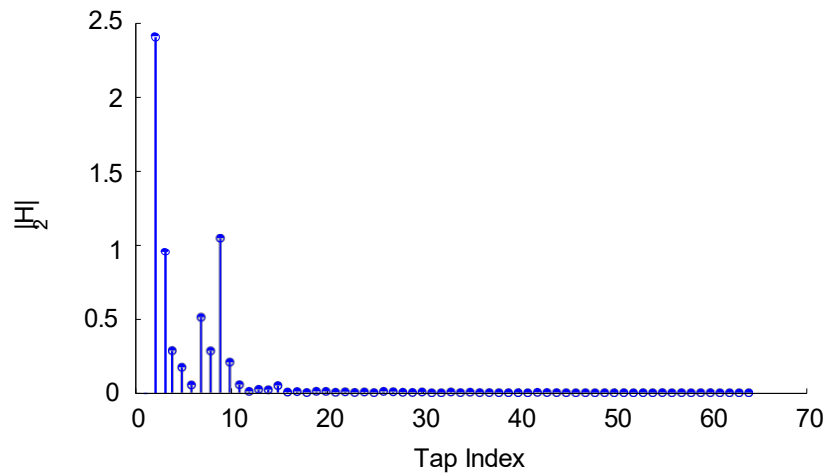
Symbols arriving along different paths sum up destructively

Problematic in
Wideband Channel!

Narrowband Channel

$h(t)$ is channel impulse response.

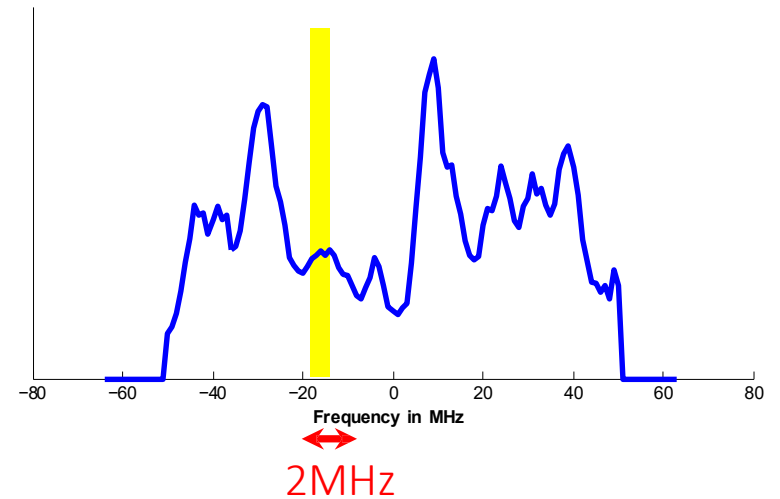
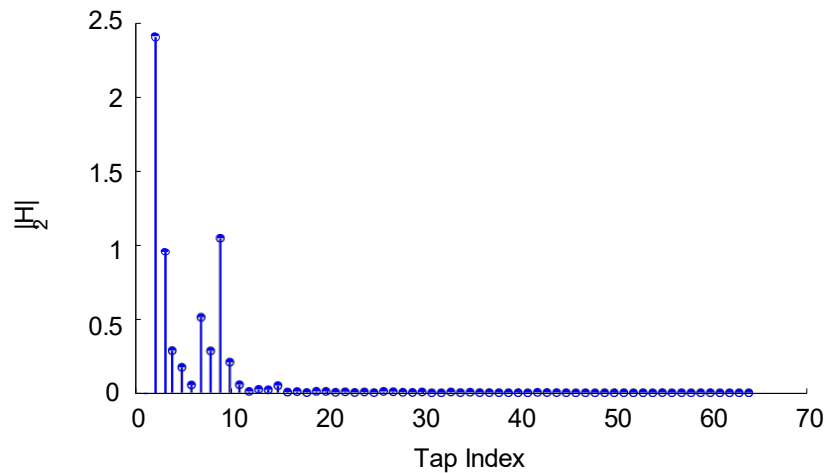
$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



Narrowband Channel

$h(t)$ is channel impulse response.

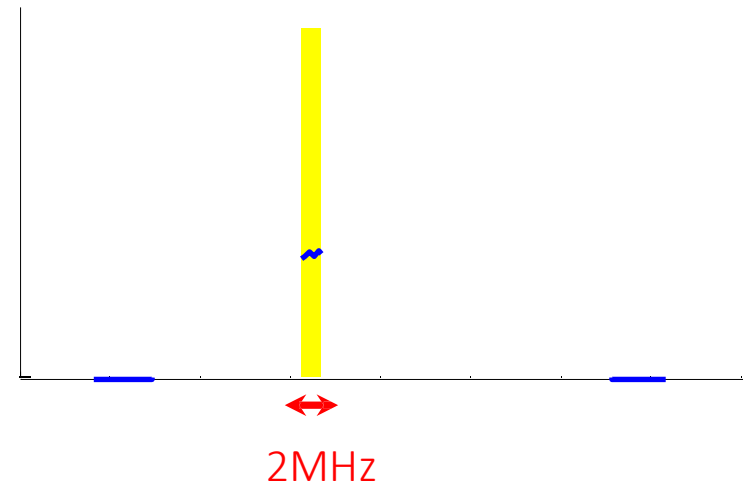
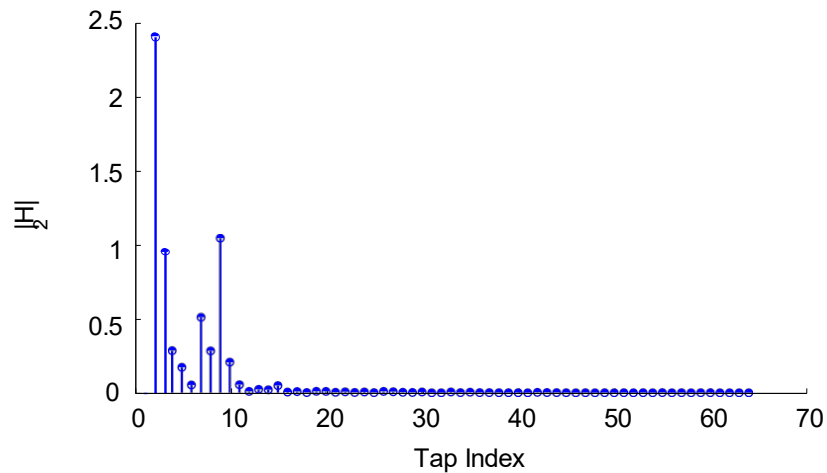
$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



Narrowband Channel

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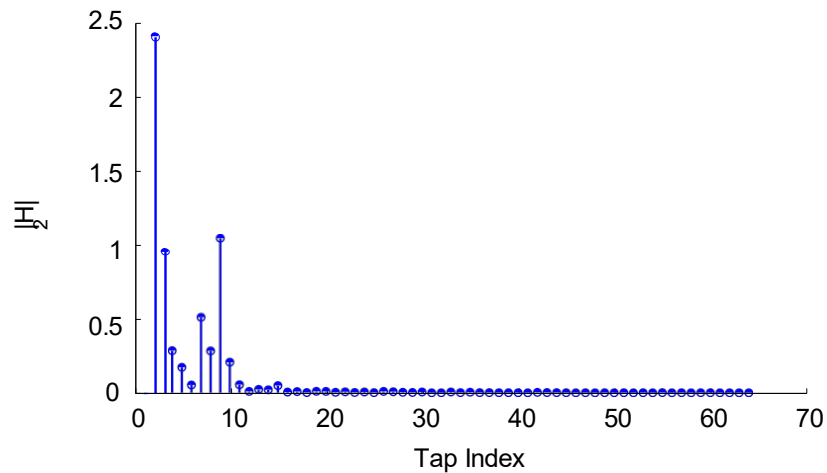
$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



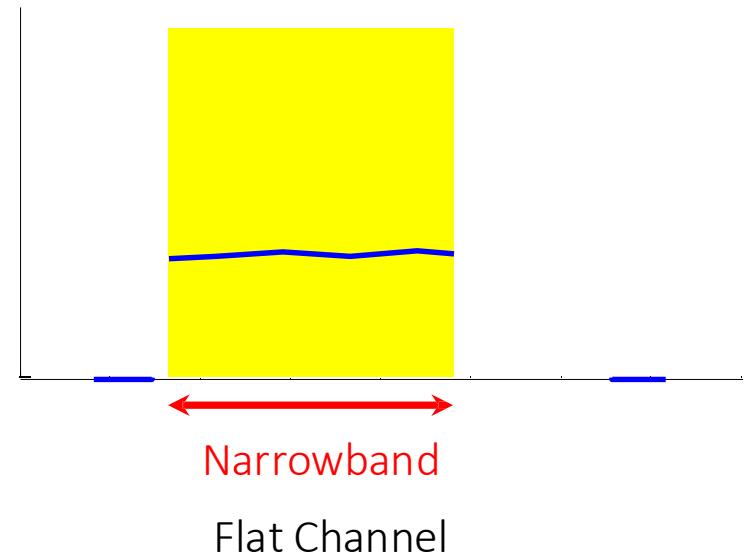
Narrowband Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



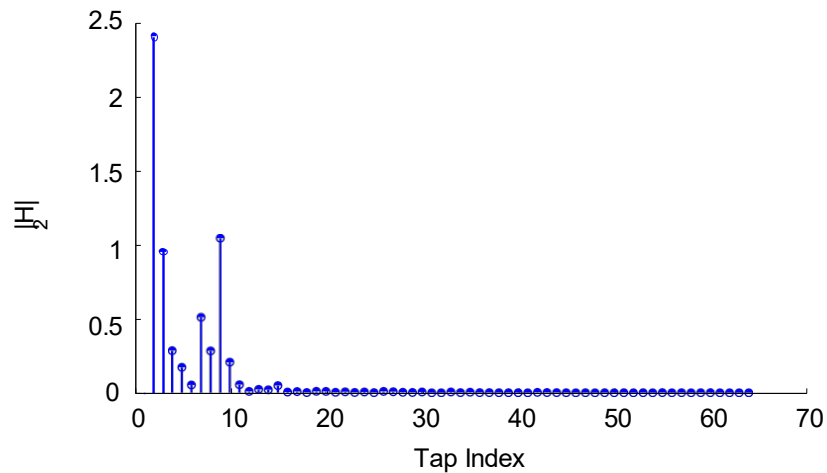
Symbol time: $T \propto \frac{1}{\text{bandwidth}}$ $\gg \tau_k$



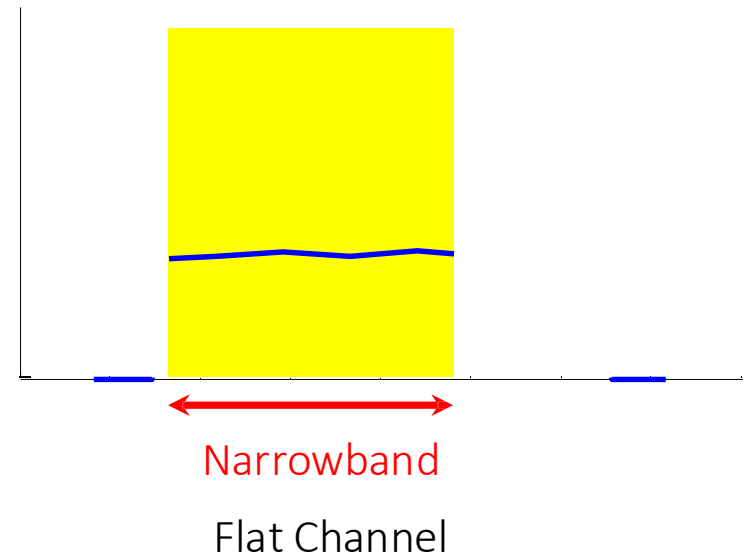
Narrowband Channel

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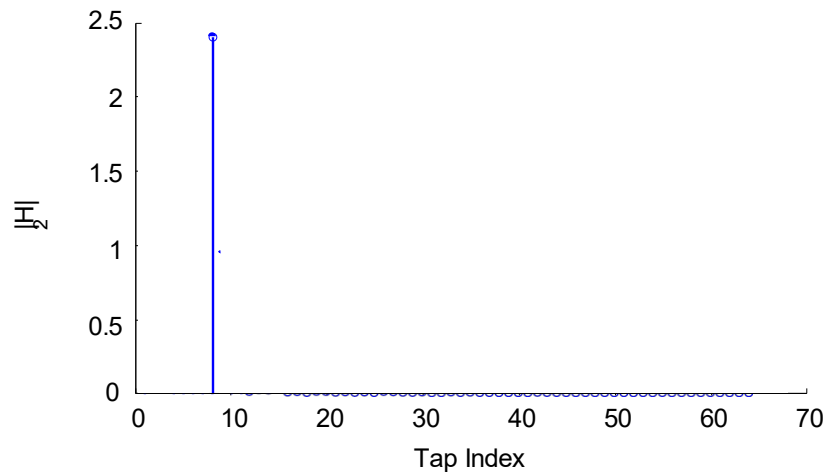
Symbol time: $T \propto \frac{1}{\text{bandwidth}}$ $\gg \tau_k$



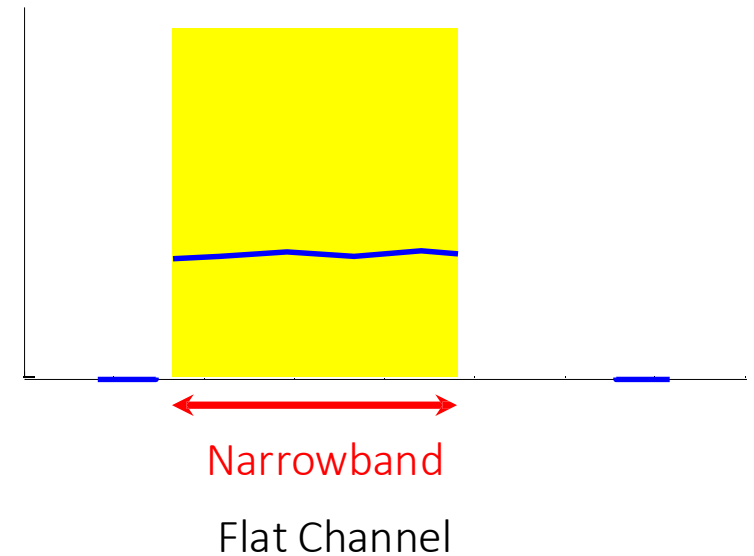
Narrowband Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



Symbol time: $T \propto \frac{1}{\text{bandwidth}}$ $\gg \tau_k$



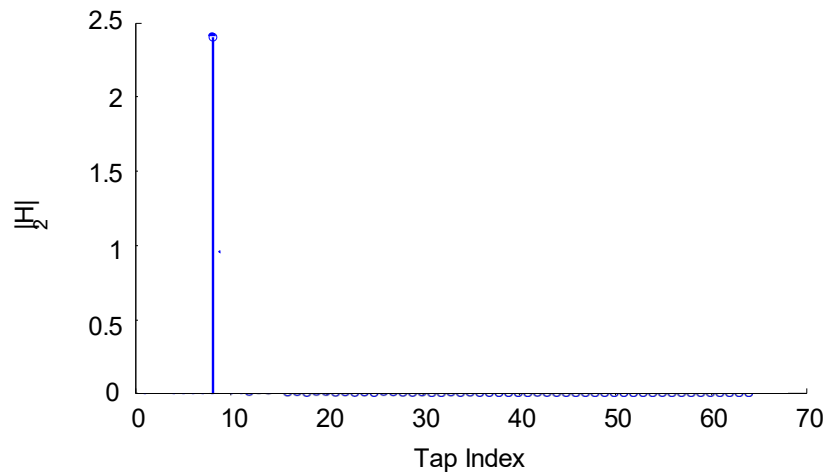
$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) \approx \sum_k h(\tau_k) x(t) = \left(\sum_k h(\tau_k) \right) x(t) = hx(t)$$

Narrowband Channel is Approximated by a Single Tap Channel

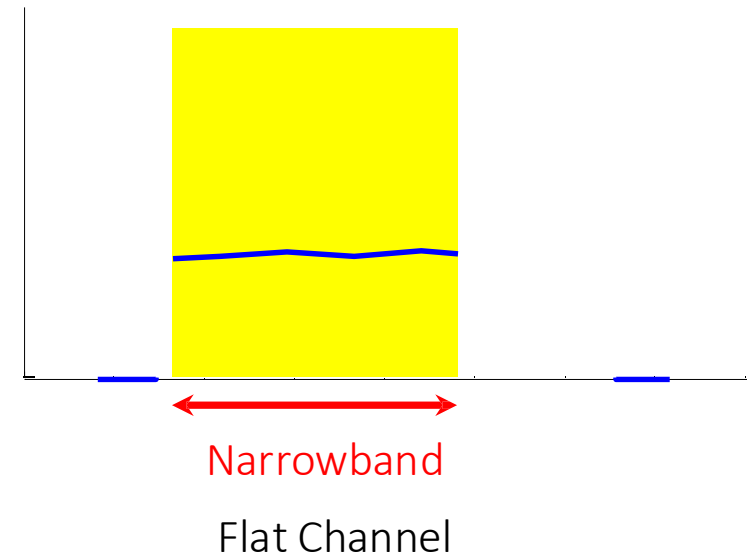
Narrowband Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t) = h x(t) \quad \Leftrightarrow \quad h X(f)$$



Symbol time: $T \propto \frac{1}{\text{bandwidth}}$ $\gg \tau_k$



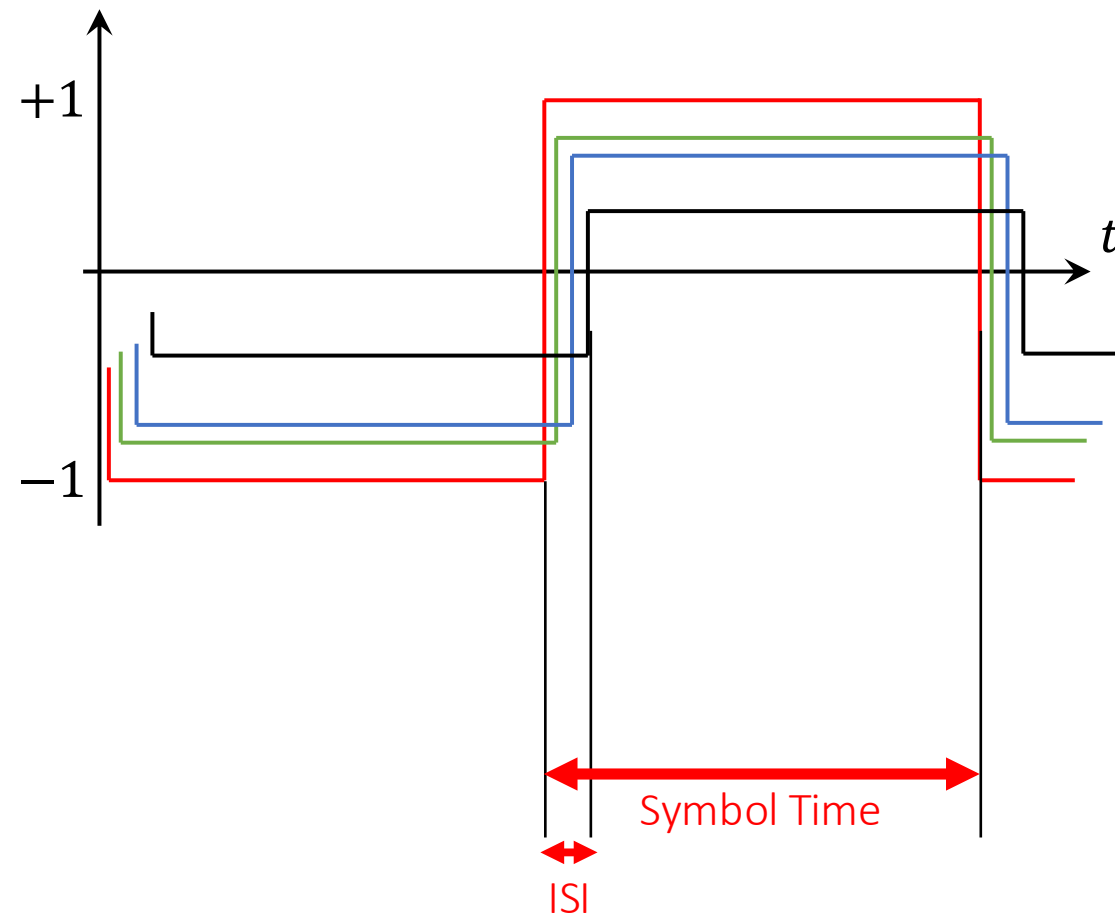
$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) \approx \sum_k h(\tau_k) x(t) = \left(\sum_k h(\tau_k) \right) x(t) = h x(t)$$

Narrowband Channel is Approximated by a Single Tap Channel

Narrowband vs. Wideband Channel

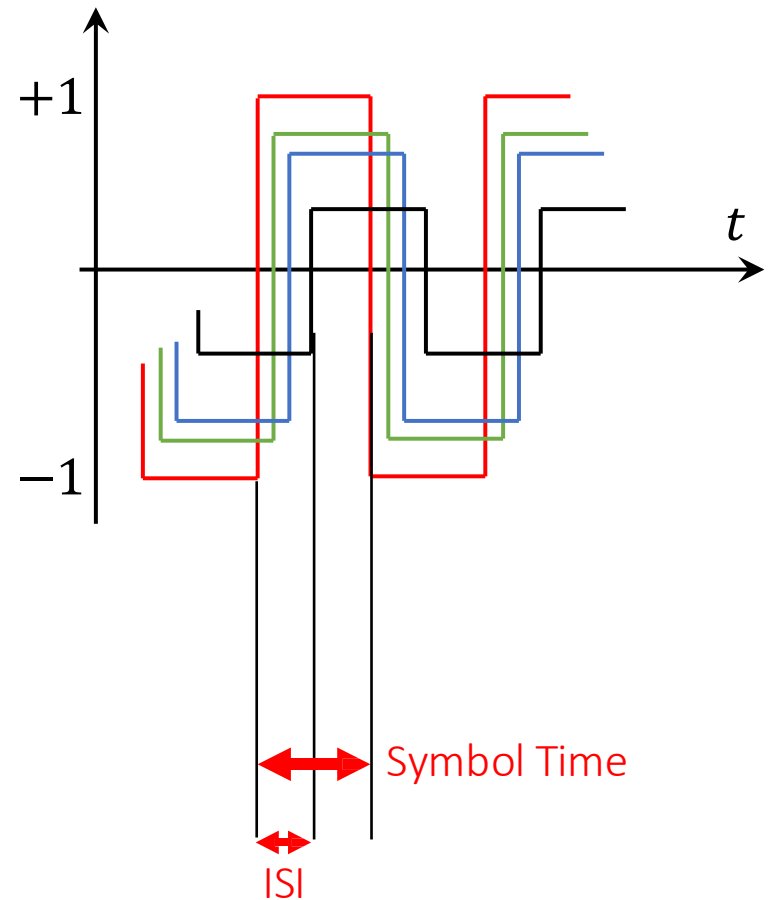
Narrowband Channel

$$h x(t)$$



Wideband Channel

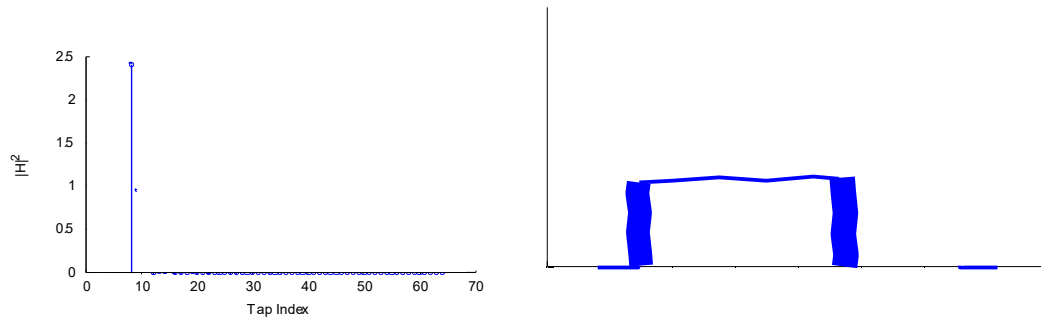
$$h(t) * x(t)$$



Narrowband vs. Wideband Channel

Narrowband Channel

$$h x(t) \iff h X(f)$$

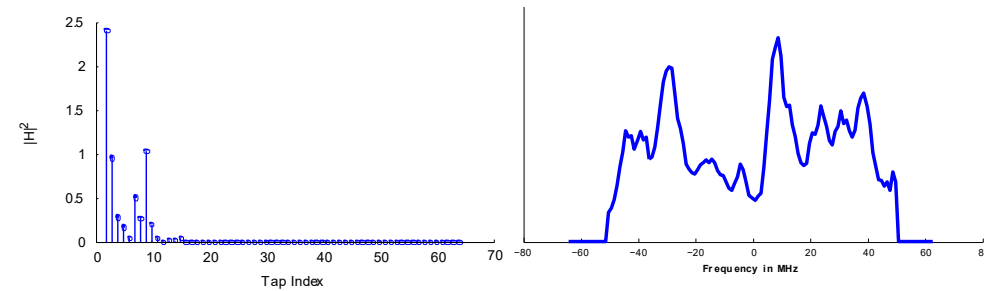


≈ Single Tap

≈ Flat Channel

Wideband Channel

$$h(t) * x(t) \iff H(f) X(f)$$



Multi-tap
Channel
(ISI)

Frequency
Selective Channel

Need to correct for ISI to
be able to decode
correctly!

Estimating and Correcting for multi-tap channel is hard
Simplify processing using: OFDM